

CHAPTER – 4

APPLICATIONS OF FIRST ORDER LINEAR DIFFERENTIAL EQUATIONS (Mathematical Modelling – I)

4.1 General

A large variety of scientific and engineering problems, wherein some type of change proceeds at a variable rate in accordance with a known law, when formulated mathematically, lead to differential equations. Therefore, the differential equations so obtained are translation of some physical laws. It is also known as **mathematical modelling** of such problems. While dealing with such problems one has to follow **three identifiable steps** which are given as under:

- a. **Translation of the physical situation into mathematical terms.** It leads to one or more differential equations, while making some assumptions consistent with the observed phenomenon. It is to be kept in mind that mathematical equations are only approximate description of the actual process because they are based on observations which are themselves approximations.
- b. **Solution of the differential equation(s), if possible.** If the equations so formulated are too hard to be solved, one may incorporate further approximations at this stage to make the problem mathematically tractable. Any such approximation must reflect the essential features of the physical process under consideration.
- c. **Interpretation of the solution in terms of the context in which the problem arose.** In particular, it must be ensured that the solution appears physically reasonable, which is unique, and depends continuously on the data of the problem.

In this chapter we shall discuss problems that give rise to linear differential equations of first order.

4.2 Exponential Growth and Decay

When a growth process is characterized by a **constant percent increase in value**, it is referred to as **exponential growth process**. When a decay process is characterized by a constant percent decline in value, it is called **exponential decay process**.

In many natural processes, the rate at which something grows or decays, depends upon its current value. For instance, in biology it has been observed that the growth of many types of bacteria is proportional to the bacteria present at that time. Similarly for various radioactive elements like Uranium-238, the decay is proportional to the amount of such an element at a particular time.

Mathematically, if the growth (or positive rate of change) of a substance is proportional to its present value, $y(t)$, at time t , it is represented as $dy/dt \propto y$, i.e.,

$$\frac{dy}{dt} = ky, \quad (4.2.1)$$

where k is a positive constant of proportionality. Its solution is given as,

$$y = c e^{kt},$$

where, c is an arbitrary constant. It represents **exponential growth**. (Fig. 4.2.1)

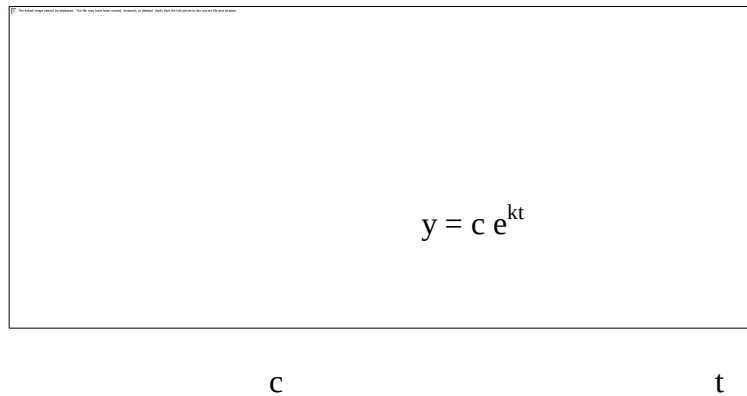


Fig. 4.2.1: Exponential Growth

Similarly, if the decay (i.e., negative rate of change) of a substance is proportional to its current value, $y(t)$, at time t , it is given as:

$$\frac{dy}{dt} = -k y, \quad (4.2.2)$$

where k is again a positive constant of proportionality. Its solution is given as,

$$y = c e^{-kt},$$

where, c is again an arbitrary constant. It represents **exponential decay** (Fig. 4.2.2).

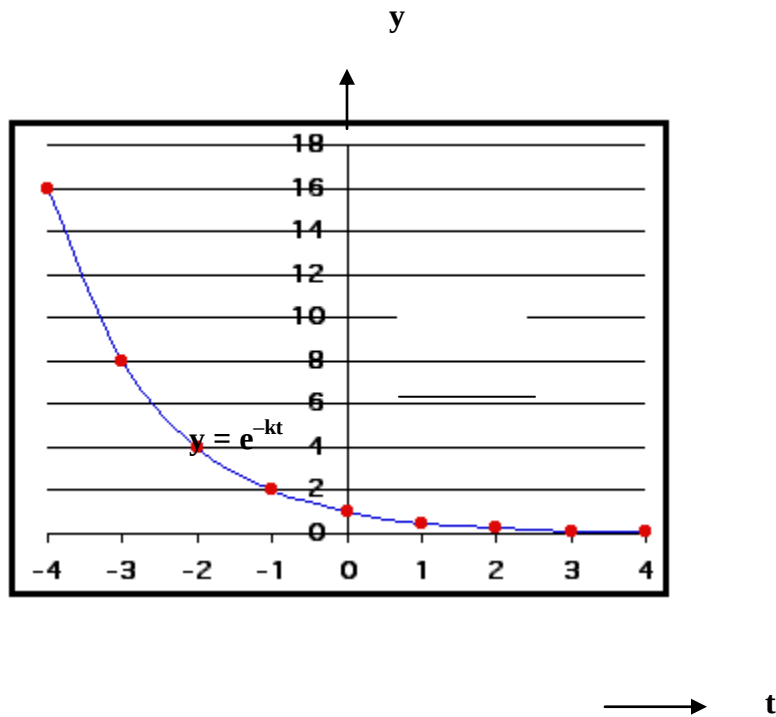


Fig. 4.2.2: Exponential Decay $y = ce^{-kt}$

It may be seen that Eqns.(4.2.1) and (4.2.2) are **first order linear differential equations of “separation of variables” form**, which may be solved as discussed earlier in Chapter 3. In this section we shall consider some interesting examples of exponential growth and decay.

It may be appreciated that in a study of growth of a population (whether human, animal, or bacterial), the function which counts the number $y(t)$ of individuals at time t is necessarily a discreet, **step function**, taking on only integer values. Therefore, the true *rate of growth* dy/dt is zero (when t lies in an open interval where y is constant), or else the derivative dy/dt does not exist (when y jumps from one integer to another). Nevertheless, useful information can often be obtained if we assume that the population $y(t)$ is a continuous function of t with a continuous derivative dy/dt at each instant. We then postulate various *laws of growth* for the population depending on the factors in the environment which may stimulate or hinder growth. For example, if the environment has little or no effect, it seems reasonable to assume that the rate of growth (or decay) is proportional to the amount present. The simplest laws of growth (or decay), as mentioned above take the form of Eqn. (4.2.1), (or (4.2.2)), where k is a constant that depends on the particular kind of population. Conditions may develop which cause the factor k to change with time in these laws of growth (or decay). As such, for instance, the growth law given by Eqn. (4.2.1) may be generalised as follows:

$$\frac{dy}{dt} = k(t) y. \quad (4.2.3)$$

If for some reason, the population cannot exceed a certain maximum M (for example, because the food supply may be exhausted), we may reasonably suppose that the rate of growth is jointly proportional to both y and $(M - y)$. Thus we have a second type of growth law as,

$$\frac{dy}{dt} = k y (M - y), \quad (4.2.4)$$

where k may be a constant or, more generally, k may change with time t . Technological improvements may tend to increase or decrease the value of M slowly, and hence we can generalise Eqn.(4.2.4) even further by allowing M to change with time.

Example 4.2.1 (Exponential Growth)

Suppose initially (i.e., at $t = 0$) the population of a bacteria is y_0 , and at $t = 1$ hour the population of bacteria is found to be $1.5 y_0$. If the rate of growth of bacteria is proportional to the current population of bacteria, determine the time required for the bacteria to double.

Solution

The given problem can be represented mathematically as:

$$\frac{dy}{dt} = ky, \quad y(0) = y_0, y(1) = 1.5 y_0. \quad (4.2.5)$$

This differential equation is of **separation of variables** form: $\frac{dy}{y} = k dt$. On integrating it gives:

$$y(t) = c e^{kt}.$$

At $t = 0$, it gives $y_0 = c e^0 = c$ (since $y(0) = y_0$). Thus we get the solution as :

$$y(t) = y_0 e^{kt} \quad (4.2.6)$$

Now, at $t = 1$, $y = 1.5 y_0$. Thus Eqn. (4.2.6) gives:

$$1.5 y_0 = y_0 e^k, \text{ or, } e^k = 1.5.$$

Taking log-natural of both sides it gives:

$$k = \ln 1.5 = 0.4054651.$$

Thus Eqn. (4.2.6) becomes:

$$y(t) = y_0 e^{0.4054651 t}, \quad (4.2.7)$$

Next we find the time required by the bacteria population to double, i.e., we have to find t when $y = 2y_0$. Therefore, in this case we shall have $y/y_0 = 2$. Thus the above equation gives:

$$2 = e^{0.4054651 t}.$$

Again taking log-natural on both sides we get:

$$t = \ln 2 / 0.4054651 \approx 1.71 \text{ hours.}$$

Thus the population of the bacteria under consideration would double in about 1.71 hours.

Note on Half Life.

The stability of a radioactive element is measured from its half-life. This is the time required by one-half of the atoms of a give quantity of radioactive element to transform into another element through disintegration or transmutation. An element with a longer half-life is more stable. For instance, the half-life of Radium isotope 226, (i.e., ${}_{88}\text{Ra}^{226}$) is about 1600 years. In this time one-half of a given quantity of Ra-226 is transmuted into Radon isotope 222 (i.e., ${}_{86}\text{Rn}^{222}$). The most commonly occurring Uranium isotope 238 (i.e., ${}_{92}\text{U}^{238}$) has a half life of about 4.47 billion (i.e, 4.47×10^9) years. In this duration half of a given quantity of ${}_{92}\text{U}^{238}$ is converted into Lead isotope 206 (i.e., ${}_{82}\text{Pb}^{206}$).

Example 4.2.2 (Exponential Decay)

In a certain radioactive substance the rate of decrease in mass is proportional to the current mass. If the mass present is reduced by half, in half an hour, what percent of the original mass is expected to remain present at the end of 0.9 hours?

Solution

Let the mass at time t is $y(t)$. Then the given problem can be represented by

$$\frac{dy}{dt} = -ky, \quad (4.2.8)$$

where k is the constant of proportionality. On solving it we get,

$$y = c e^{-kt}, \quad (4.2.9)$$

where c is the constant of integration. If y_0 is the mass at $t = 0$, the above equation becomes,

$$y_0 = c e^0 = c. \quad (4.2.10)$$

Since the mass is reduced by half, in half an hour, so for $t = \frac{1}{2}$, Eqn. (4.2.9) gives,

$$\frac{1}{2} y_0 = c e^{-k/2} = y_0 e^{-k/2}.$$

Thus,

$$\frac{1}{2} = e^{-k/2},$$

Taking log-natural of both sides we get $\frac{1}{2} k = -\ln(1/2) = \ln(2)$, or,

$$k = \ln 4.$$

Therefore, the desired solution is,

$$y = y_0 e^{-(\ln 4)t} . \quad (4.2.11)$$

This solution can be put as: $y = y_0/e^{t(\ln 4)} = y_0/4^t$. Therefore, when $t = 0.9$ (given), then the above equation gives:

$$y = y_0/4^{0.9} = 0.2871745 y_0 = 29 \% \text{ of } y_0 \text{ (approx.)}$$

Thus at the end of 0.9 hours about 29 percent of the original quantity of the given radioactive substance would be left.

Example 4.2.3 (Exponential Decay)

A nuclear reactor converts relatively stable Uranium-238 (i.e., ${}_{92}\text{U}^{238}$) into the isotope Plutonium-239 (i.e., ${}_{94}\text{Pu}^{239}$). After 15 years it is determined that 0.043 % of the initial amount, y_0 , of Plutonium has disintegrated or transmuted. Find the half-life of this isotope if the rate of disintegration is proportional to the amount remaining.

Solution

Let $y(t)$ denote the amount of Plutonium remaining at any time t , and let $y(0) = y_0$. Then the given problem can be expressed mathematically as:

$$\frac{dy}{dt} = -ky, \quad y(0) = y_0.$$

On solving this differential equation (variables separable form) we get:

$$y(t) = y_0 e^{-kt} . \quad (4.2.12)$$

It is given that 0.043 % of the atoms of y_0 have disintegrated in 15 years. So at $t = 15$ years, 99.957 % of the substance remains. Thus we have,

$$y(15) = 0.99957 y_0.$$

Therefore, for $t = 15$ years, the above equation gives:

$$0.99957 y_0 = y_0 e^{-15k}, \text{ or } 0.99957 = e^{-15k} .$$

Thus on taking log natural on both sides of this equation we get,

$$-15 k = \ln (0.99957), \text{ or,}$$

$$k = \frac{\ln(0.99957)}{-15} = 2.8673 \times 10^{-5} = 0.000028673.$$

Thus the solution given by Eqn. (4.2.12) becomes,

$$y(t) = y_0 e^{-0.000028673 t} \quad (4.2.13)$$

Now, the half-life is the corresponding value of time for which $y(t) = y_0/2$. Thus the above equation gives,

$$y_0/2 = y_0 e^{-0.000028673 t}, \text{ or } 0.5 = e^{-0.000028673 t}.$$

It gives,

$$-0.000028673 t = \ln (0.5) .$$

Thus we get,

$$t = \ln 2 / 0.000028673 \approx 24236 \text{ years,}$$

which is the required half-life of the Plutonium isotope-239.

Example 4.2.4 (Exponential Decay)

A radioactive substance disintegrates at a rate proportional to the amount of the substance present. If 100 units of this substance disintegrate to 50 units in 1700 years, how many of them will remain after a total of 2500 years?

Solution

Let $y(t)$ denote the amount of the substance at any time t , and let $y(0) = y_0$. Then the given problem can be expressed mathematically as $dy/dt = -ky$. Its solution is,

$$y(t) = y_0 e^{-kt}, \quad (4.2.14)$$

where, $y(0) = y_0 = 100$, and $y(1700) = 50$, and, we have to determine $y(2500)$. Using the condition $y_0 = 100$ (given), and $y(1700) = 50$ (given), the above equation gives,

$$50 = 100 e^{-1700 k}.$$

Therefore, taking log-natural on both sides of this equation we get,

$$\begin{aligned} -1700 k &= \ln(1/2) = -\ln(2), \\ \text{i.e.,} \\ k &= 0.0004077. \end{aligned}$$

Thus Eqn. (4.2.14) becomes,

$$y(t) = 100 e^{-0.0004077 t}.$$

Therefore, for $t = 2500$ years, the number of the remaining units would be,

$$y(2500) = 100 e^{-1.019} \approx 35 \text{ units..}$$

Example 4.2.5 - Lambert's Law of Absorption (Exponential Decay).

This law was expounded by German physicist and mathematician *Johann Heinrich Lambert* (1726-1777 AD). It states that the absorption of light in a very thin transparent layer is proportional to the thickness of the layer and to the amount of light incident on that layer. Formulate this in terms of a differential equation and solve it.

Solution

Let y = amount of incident light, Δy = absorbed light, Δx = thickness of the layer. Then mathematically Lambert's law can be stated as, $\Delta y \propto y \Delta x$, i.e.,

$$\Delta y = -k y \Delta x, \text{ or } \Delta y/\Delta x = -ky,$$

where k is the constant of proportionality. Letting $\Delta x \rightarrow 0$ it becomes,

$$dy/dx = -ky, \text{ or } dy/y = -k dx.$$

On integration it yields:

$$y(x) = c e^{-k x}.$$

It gives the amount of light in a thick layer at depth x from the surface of incidence.

4.3 Radioactive Carbon Dating

In the atmosphere, the radioactive Carbon isotope ${}^6\text{C}^{14}$ is produced due to the effect of cosmic radiation on Nitrogen. It has been found that the ratio of the amount of ${}^6\text{C}^{14}$ to ordinary Carbon ${}^6\text{C}^{12}$ in the atmosphere is constant. Accordingly the ratio of these Carbon isotopes in the living organisms is also the same as that in the atmosphere. When an organism dies, the absorption of ${}^6\text{C}^{14}$ by breathing and eating stops. Thus one can estimate the age of a fossil by comparing the proportionate amount of ${}^6\text{C}^{14}$ present in the fossil with the constant ratio found in the atmosphere. This method is based on the knowledge that half-age of the radioactive Carbon isotope ${}^6\text{C}^{14}$ is 5730 years.

The above mentioned theory was put forward by a chemist **Willard Libby** in **1940**. For that he was awarded **Nobel Prize in Chemistry in 1960**. This method has been in use ever since for estimating the ages of various ancient items like estimation of the ages of wooden furniture in Egyptian tombs, woven flax wrappings of Dead Sea scrolls, etc.

Example 4.3.1 (Carbon Dating)

A fossil bone is found to contain $1/1000$ the original amount of ${}^6\text{C}^{14}$. Determine the age of the fossil.

Solution

The given phenomenon is governed by the following equation, which gives the amount of the radioactive substance at any time t ,

$$y(t) = y_0 e^{-kt},$$

where y_0 is the initial amount of the radioactive substance. For $t = 5730$ years (the half-age of ${}^6\text{C}^{14}$), we have $y(t) = y_0/2$. Thus we can determine the value of the constant k from the above equation as under,

$$y_0/2 = y_0 e^{-5730 k}, \text{ or } -5730 k = \ln(1/2).$$

Thus we get,

$$k = \frac{\ln(1/2)}{-5730} = 0.000120$$

Therefore the above equation becomes,

$$y(t) = y_0 e^{-0.00012097 t}.$$

Thus, when $y(t) = y_0/1000$, as given, we get,

$$y_0/1000 = y_0 e^{-0.00012097 t}.$$

On taking log-natural of both sides this equation gives,

$$-0.00012097 t = \ln(1/1000) = -\ln(1000).$$

Therefore,

$$t = \ln(1000) / 0.00012097 = 57136 \text{ years.}$$

Note.

In the above method radioactive Carbon dating much of the accuracy of results depends upon the chemical analysis of the fossil. In order to obtain better estimations of ${}^{14}_6\text{C}$ present in the fossil, destruction of large samples of the specimen are required. The same may not always be possible from archival point of view. In view of the same the age estimation of the fossil in the above example is not very accurate.

In the recent years geologists have shown that age estimation of the fossils by the above mentioned method may be out by as much as 3500 years in certain cases. One of the possible reasons for this error is the fact that ${}^{14}_6\text{C}$ levels in the air vary with time. They have devised another method for the purpose, based on the fact that the living organisms ingest traces of Uranium. By measuring the relative amounts of Uranium and Thorium (the isotope into which Uranium decays), and by knowing the half-lives of these elements, one can determine the age of the fossil. By this method one can estimate the ages of even 500,000 years old fossils. However, this method is not applicable to marine fossils. Some other techniques can also be used for the purpose including the use of Potassium-40 and Argon-40 (which can estimate ages up to millions of years), and non-isotopic methods based on the use of amino acids.

EXERCISE - 4.1

(Exponential Growth & Decay, Carbon Dating “*Separation of Variables*” Form)

Note: Answers are given against the respective questions within square brackets.

1. In a culture of yeast the rate of growth dy/dt is proportional to its population present at any time t , i.e., $dy/dt = ky$. Verify that $y(t) = y_0 e^{kt}$, where $y_0 = y(0)$. If y doubles in 1 day, how much can be expected after 1 week at the same rate of growth? How much after 2 weeks?
 $[y(7) = y_0 2^7 = 128 y_0, y(14) = y_0 2^{14} = 16384 y_0]$
2. In a culture of yeast the amount of active yeast grows at a rate proportional to the amount present. If that amount doubles in 2 hours, how much can be anticipated at the end of 5 hours 30 minutes?
 $[y(t) = 2^{2.75} y_0 \approx 6.73 y_0]$
3. The population of a certain community is known to increase at a rate proportional to the number of people present at any time. If the population has doubled in 5 years, how long will it take to triple? to quadruple?
 $[7.9 \text{ years}; 10 \text{ years}]$
4. If the population in problem 2 is 10,000 after 3 years, what was the initial population? What will be the population in 10 years?
 $[3536, 113137]$
5. The population of a town grows at a rate proportional to the population at any time. Its initial population of 500 increases by 15% in 10 years. What will be the population in 30 years?
 $[760]$
6. The number of bacteria in a culture increases at a rate proportional to the number of bacteria present at any time. After 3 hours it is observed that there are 400 bacteria present. After 10 hours there are 2000 bacteria present. What is the initial number of bacteria?
 $[200]$
7. Express y as a function of t for each of the growth laws in Eqns.(4.2.1) and (4.2.4) (with k and M both constants). Show that the result for (4.2.4) can be expressed as,

$$y(t) = \frac{M e^{\alpha(t-t_0)}}{1 + e^{\alpha(t-t_0)}} = \frac{M}{1 + e^{-\alpha(t-t_0)}},$$

where α is a constant and t_0 is the time at which $y = M/2$.

$$[\text{For Eqn (4.2.1), } y(t) = y_0 e^{k(t-t_0)} = \frac{M}{2} e^{k(t-t_0)}; \text{ for Eqn (4.2.4), } \alpha = M k]$$

8. The radioactive isotope of lead, ${}_{82}\text{Pb}^{209}$, decays at a rate proportional to the amount present at any time and has a half-life of 3.3 hours. If 1 gram of lead is present initially, how long will it take for 90% of the lead to decay? **[11 hours]**

9. Initially there were 100 milligrams of a radioactive substance present. After 6 hours the mass decreased by 3%. If the rate of decay is proportional to the amount of substance present at any time, find the amount remaining after 24 hours.

[88.5 mg approx.]

10. Determine the half-life of the substance mentioned in problem 9. **[136.5 hours]**

11. Show that, in general, the half-life of a radioactive element is given by the relation,

$$t = \frac{(t_2 - t_1) \ln 2}{\ln(y_1/y_2)}, \text{ where, } y_1 = y(t_1), \text{ and } y_2 = y(t_2), t_1 < t_2.$$

12. A substance decreases at a yearly rate which is numerically double the number of units present. If there are initially 1000 units, how many will remain at the end of 1 year?

[1000 e^{-2} = 135.3 units]

13. A quantity of Radium, ${}_{88}\text{R}^{226}$, disintegrates at a rate proportional to the amount present. Half of the initial amount is left at the end of 1700 years. How long did it take for one quarter to disintegrate? **[1700 $\ln(4/3) / \ln(2)$ = 705 years]**

14. The half-life for Radium, ${}_{88}\text{R}^{226}$, is approximately 1600 years. Find what percentage of a given quantity of Radium disintegrates in 100 years. **[100(1 - 2^{-1/16})% = 4.2%]**

15. After 1000 years there were 80 units of a radioactive substance remaining, while after 2000 years there remained 64 units. If the substance disintegrates at a rate proportional to the amount present, what was the initial amount? **[100 units]**

16. Derive a formula for the amount of a substance, which disintegrates at a rate proportional to the current amount, present at the end of t years if y_0 units disintegrate to y_1 in t_1 years.

$$[y(t) = y_0 \left(\frac{y_1}{y_0} \right)^{t/t_1}]$$

17. Observations show that the rate of change of atmospheric pressure $y(x)$, with altitude x is proportional to the pressure. Assuming that the pressure at 6000 meters (about 18000 feet) is half its value $y(0) = y_0$ at the sea level, derive the formula for the pressure at any height.

$$[y(x) = y_0 e^{-0.000116 x}]$$

18. A bone is claimed to be 2000 years old. Determine the quantity of ${}^6\text{C}^{14}$ present in it.
[78.5% of y_0]
19. When a vertical beam of light passes through a transparent substance, the rate at which its intensity I decreases is proportional to $I(t)$, where t represents the thickness of the medium in feet. In clear sea-water the intensity 3 feet below the surface is 25% of the initial intensity I_0 of the incident vertical beam. Calculate the intensity of the beam 15 feet below the surface.
[$I(15) = 0.00098 I_0 \approx 0.1\%$ of I_0]

4.4 Newton's Law of Cooling

This law states that the rate at which the temperature of a body changes is proportional to the difference between its temperature T and the temperature T_0 of the surrounding medium. Mathematically it is stated as,

$$\frac{dT}{dt} = -k(T - T_0), \quad (4.4.1)$$

where k is a positive constant of proportionality. The negative sign in the above equation indicates that rate of change of temperature T will decrease with time. This law helps us in many ways as we shall see in the following examples.

Example 4.4.1

Suppose a body cools from 200° to 100° in 40 minutes while immersed in a medium whose temperature is kept constant, say $T_0 = 10^\circ$. Calculate the value of k in Eqn (4.4.1).

Solution

For the given data, we can separate the variables in Eqn (4.4.1) to write,

$$\int_{200}^{100} \frac{dT}{T - 10} = -k \int_0^{40} dt.$$

Here we have inserted specific limits of integration instead of using indefinite integrals with an arbitrary constant. On integrating we get,

$$\ln 90 - \ln 190 = -40k, \text{ i.e., } k = \frac{1}{40} \ln \frac{19}{9} = 0.0186803$$

Example 4.4.2

In the above example calculate the time required by the same material to cool from 200° to 100° if the temperature of the medium is kept at 5° .

Solution

In this case denoting the unknown time by x we have,

$$\int_{200}^{100} \frac{dT}{T-5} = -k \int_0^x dt,$$

i.e.,

$$\ln(95/195) = -kx = -0.0186803 x.$$

Thus we get,

$$x = 38.5 \text{ minutes,}$$

which is the time required by the material to cool from 200° to 100° .

Example 4.4.3

In the above example find the time required to cool the same substance from 100° to 10° with the medium kept at 5° .

Solution

In this case we have,
$$\int_{100}^{10} \frac{dT}{T-5} = -k \int_0^x dt,$$

i.e.,

$$\ln(5/95) = -kx = -0.0186803 x.$$

Thus we get the required time as, $x = 158$ minutes (approx.)

Note

In the above examples we see that rate of cooling decreases considerably as the temperature of the body begins to approach the temperature of the medium. The temperature drop from 100° to 10° takes more than four times as long as the change from 200° to 100°

Example 4.4.4

An object is heated to a temperature of 100° . Then at $t = 0$ it is placed in water having a temperature of 30° . At the end of 3 minutes the temperature of the object reduces to 70° . Find the time taken by the object to cool down to 31° .

Solution

Using Newton's law of cooling, the given problem can be expressed mathematically as,

$$\frac{dT}{dt} = -k(T - 30), \quad T(0) = 100, \quad T(3) = 70.$$

It can be written as $\frac{dT}{T-30} = -kdt$, which on integration gives,

$$\ln |T-30| = -kt + \ln c = \ln ce^{-kt}.$$

Thus we get,

$$T(t) = ce^{-kt} + 30.$$

Using the initial condition $T(0) = 100$, it gives $100 = ce^0 + 30$, i.e., $c = 70$. Thus we get,

$$T(t) = 70 e^{-kt} + 30.$$

To determine k we use the given condition: $T(3) = 70$ in the above equation to get,

$$70 = 70 e^{-3k} + 30, \text{ i.e., } e^{-3k} = 4/7.$$

Taking log-natural of both sides we get, $k = [\ln (4/7)]/(-3) = 0.18654$. Thus we get the solution of the given differential equation as,

$$T(t) = 70 e^{-0.18654 t} + 30.$$

In order to find the time taken by the given object to cool down to 31° , we put $T(t) = 31$ in the above equation to get,

$$31 = 70 e^{-0.18654 t} + 30, \text{ i.e., } e^{-0.18654 t} = 1/70.$$

Taking log-natural on both sides it gives,

$$t = [\ln (1/70)]/(-0.18654) = 22.78 .$$

Therefore, the given object will cool down to 31° after about 23 minutes.

Example 4.4.5

When a cake is removed from a baking oven, its temperature is 300° F. Three minutes later its temperature is found to be 200° F. If the room temperature is 70° F, how long will it take the cake to cool down nearly to the room temperature, say to 70.1° F?

Solution

The give problem can be expressed mathematically as,

$$\frac{dT}{dt} = -k(T-70), T(0) = 300, T(3) = 200.$$

This differential equation is of variable separable form which can be written as,

$$\frac{dT}{T-70} = -k dt.$$

On integrating it gives,

$$\ln |T-70| = -kt + \ln c = \ln ce^{-kt}.$$

Thus we get,

$$T(t) = ce^{-kt} + 70.$$

For $T(0) = 300$ it gives,

$$300 = ce^0 + 70 = c + 70, \text{ i.e., } c = 230.$$

Thus the above equation becomes,

$$T(t) = 230 e^{-kt} + 70.$$

Again, for $T(3) = 200$ this equation gives, $200 = 230 e^{-3k} + 70$, i.e., $e^{-3k} = 13/23$. Taking log-natural on both sides it gives,

$$k = [\ln (13/23)]/(-3) = 0.190182.$$

Thus the above equation becomes,

$$T(t) = 230 e^{-0.190182 t} + 70.$$

In order to find the time taken by the cake to cool down to 70.1°F , we put $T(t)=70.1$ in the above equation to get,

$$70.1 = 230 e^{-0.190182 t} + 70.$$

Thus,

$$-0.190182 t = \ln (0.1/230).$$

Thus we get,

$$t = 41 \text{ minutes (approx.)}.$$

Therefore, the cake would cool down to nearly the room temperature in about 41 minutes.

4.5 Heating and Cooling of Buildings

Man has ever been fighting against the weather hazards to keep his life comfortable. During summer one needs to keep his accommodation cold while during winter some heating arrangement is required for the purpose. In order to tackle these problems we can formulate a mathematical model to take care of 24-hour temperature profile inside a building as a function of the following phenomena:

- a. Temperature of the body itself, $T(t)$;
- b. The outside temperature, $M(t)$;
- c. The heat generated inside the building, $H(t)$;

- d. The furnace heating inside the building
 e. Cooling due to air conditioning inside the building } $U(t)$

In order to build up a mathematical model for the temperature inside a building we use compartmental analysis. Consider a building as a single compartment and suppose that $T(t)$ is the temperature inside the building at time t . Then the rate of change of temperature in the building is determined by all the factors that generate or dissipate heat. In this regard we consider three main factors as given below:

a. Heat produced by the people, light, and machines inside the building.

This causes a rate of increase in the temperature of the building. We denote it by $H(t)$, and in general it is described in terms of *energy per unit time* (such as British thermal units per hour).

b. The heating (or cooling) supplied by the furnace/heater (or air conditioner)

It causes a rate of increase (or decrease) in the temperature of the building. We represent it by $U(t)$. In general it is also described in terms of *energy per unit time* (such as British thermal units per hour).

Note.

In general, the additional heating rate $H(t)$ and the furnace (or air conditioning) rate $U(t)$ are described in terms of *energy per unit time* (such as British thermal units per hour). However, by multiplying by the heat capacity of the building (in terms of degrees *temperature per unit heat energy*), we can express the two quantities $H(t)$ and $U(t)$ in terms of temperature per unit time.

c. The effect of the outside temperature $M(t)$ on the temperature inside the building.

Experimental evidence has shown that this factor can be modelled using *Newton's law of Cooling*. In the present situation this law can be expressed as: "The rate of change of temperature $T(t)$ is proportional to the difference between the outside temperature $M(t)$ and the inside temperature $T(t)$ ". That is, the rate of change in the building temperature due to $M(t)$ is equal to:

$$K[M(t) - T(t)],$$

where the constant of proportionality K is positive and depends upon the physical properties of the building, such as the number of doors and windows and the type of

insulation. However, it does not depend upon M , T or t . Hence, when the outside temperature is greater than the inside temperature, then $[M(t) - T(t)] > 0$, and there is an increase in the rate of change of the building temperature due to $M(t)$. However, when the outside temperature is less than the inside temperature, then $[M(t) - T(t)] < 0$ and there is a decrease in this rate of change of the temperature.

Summarizing the above discussion we get,

$$\frac{dT}{dt} = K [M(t) - T(t)] + H(t) + U(t), \quad (4.5.1)$$

where the additional heating rate $H(t)$ is always non-negative and $U(t)$ is positive for heating due to a furnace/heater and negative for air conditioner cooling.

Note

A more detailed model of the temperature dynamics of the building could involve more variables to represent different temperatures in different rooms or zones. Such an approach would use compartmental analysis, with the rooms as different compartments.

Eqn. (4.5.1) can be written as,

$$\frac{dT}{dt} + K T(t) = K M(t) + H(t) + U(t), \quad (4.5.2)$$

which is a linear differential equation of first order, of the form: $dT/dt + P(t) T = Q(t)$, where

$P = K$, and $Q(t) = KM(t) + H(t) + U(t)$. Thus its integrating factor is:

$$e^{\int P dt} = e^{\int K dt} = e^{Kt}.$$

Therefore, using the formula, $T(t) = e^{-Pt} \left[\int e^{Pt} Q(t) dt + c \right]$, we get the solution of Eqn. (4.5.2) as,

$$T(t) = e^{-Kt} \left[\int e^{Kt} \{K M(t) + H(t) + U(t)\} dt + c \right]. \quad (4.5.3)$$

Example 4.5.1

Suppose that at the end of a day (at time t_0), when people leave a building, the outside temperature stays constant at M_0 , the additional heating rate H inside the building is zero, and the furnace/air conditioner rate U is zero. Determine $T(t)$, given the initial condition $T(t_0) = T_0$.

Solution

In this case we have, $M = M_0$, $H = 0$, and $U = 0$. Thus from Eqn (4.5.3) we get,

$$\begin{aligned} T(t) &= e^{-Kt} \left[\int e^{Kt} K M_0 dt + c \right] = e^{-Kt} [M_0 e^{Kt} + c] \\ &= M_0 + c e^{-Kt}. \end{aligned}$$

Using the initial condition $T(t_0) = T_0$ we get the value of c as: $(T_0 - M_0) e^{Kt_0}$. Hence we get the value of $T(t)$ as,

$$T(t) = M_0 + (T_0 - M_0) e^{-K(t-t_0)}. \quad (4.5.4)$$

An analysis of the result

When $M_0 < T_0$, the temperature $T(t)$ given by the solution (4.5.4) decreases exponentially from the initial temperature T_0 (of the building) to the final (outside) temperature M_0 .

To determine a measure of the time it takes for the temperature to change “substantially”, we consider a simple linear differential equation: $dA/dt = -\alpha A$. Its solution has the form: $A(t) = A(0)e^{-\alpha t}$. Now, as the time keeps on passing (i.e, as $t \rightarrow \infty$), the function $A(t)$ either decays exponentially (for $\alpha > 0$), or grows exponentially (for $\alpha < 0$). Taking $t = 1/\alpha$ it gives: $A(1/\alpha) = A(0)/e = 0.368 A(0)$. Thus in either of the above two cases, the time it takes for $A(t)$ to change from $A(0)$ to $A(0)/e$ is just $1/\alpha$. The quantity $1/|\alpha|$, which is independent of $A(0)$ is called the **time constant** for the equation. For linear equations of more general form,

$$dA/dt = -\alpha A + g(t),$$

we again refer to $1/|\alpha|$ as the time constant.

Returning to our above example, we see that the time constant is $1/K$. It represents the time it takes for the temperature difference $(T - M_0)$ to change from $(T_0 - M_0)$ to $(T_0 - M_0)/e$. We also call $1/K$ as the **time constant for the building** under consideration (without heating or air conditioning). A typical value for the time constant of a building is 2 to 4 hours, but the time constant can be much shorter if the windows/door are open, or if a fan is circulating the air. Or, it can be much longer if the building is well insulated.

Therefore, in the context of Example 4.5.1, we can use the notion of time constant to assert that:

- a. The building temperature changes exponentially with a time constant of $1/K$,
- b. The above temperature inside the building during spring and autumn is considered in the next example.

Example 4.5.2

For the building temperature $T(t)$ if the additional heating rate $H(t)$ is equal to the constant H_0 , there is no heating or cooling ($U(t) = 0$), and the outside temperature M varies as a sine wave over a 24-hour period, with its minimum at $t = 0$ (midnight) and its maximum at $t = 12$ (noon). That is,

$$M(t) = M_0 - B \cos \omega t,$$

where B is a positive constant, M_0 is the average outside temperature, and $\omega = 2\pi/24 = \pi/12$ radians/hour. (This could be the situation during the spring or autumn (fall) when there is neither furnace heating nor air conditioning.

Solution

In this case we have the differential equation,

$$\frac{dT}{dt} = K [M(t) - T(t)] + H(t) + U(t) = K[M_0 - B \cos \omega t - T(t)] + H_0 + 0$$

or,

$$\frac{dT}{dt} + K T(t) = K[M_0 - B \cos \omega t] + H_0. \quad (4.5.5)$$

It is again of the form : $dT/dt + K T(t) = Q(t)$, where, $Q(t) = K[M_0 - B \cos \omega t] + H_0$, and the integrating factor is $e^{\int K dt} = e^{Kt}$. Thus the solution of Eqn. (4.5.5) is given as,

$$\begin{aligned} T(t) &= e^{-Kt} \left[\int e^{Kt} \{K(M_0 - B \cos \omega t) + H_0\} dt + C \right] \\ &= e^{-Kt} \left[M_0 e^{Kt} - KB \frac{e^{Kt} (K \cos \omega t + \omega \sin \omega t)}{K^2 + \omega^2} + H_0 e^{Kt}/K + C \right]. \end{aligned}$$

Thus we have,

$$T(t) = (M_0 + H_0/K) - KB \frac{(K \cos \omega t + \omega \sin \omega t)}{K^2 + \omega^2} + C e^{-Kt}. \quad (4.5.6)$$

We can choose the constant C so that at midnight ($t = 0$), the value of the temperature T is equal to some initial temperature T_0 . Thus,

$$C = T_0 - (M_0 + H_0/K) + KB \frac{K}{K^2 + \omega^2}$$

An Analysis of the Result

It may be noticed that the last term in Eqn.(4.5.6) involving the constant C tends to zero exponentially. The constant term $(M_0 + H_0/K)$ in this equation represents the daily average temperature inside the building (neglecting the exponential term). When there is no additional heating rate inside the building ($H_0 = 0$), this average temperature is equal to the average outside temperature M_0 . The term $[KB \frac{(K \cos \omega t + \omega \sin \omega t)}{K^2 + \omega^2}]$ represents the

sinusoidal variation of temperature inside the building responding to the outside temperature variation. This term can be written as $[KB (K^2 + \omega^2)^{-1/2} \cos(\omega t - \phi)]$, where, $\tan \phi = \omega/K$. Therefore, the sinusoidal variation inside the building lags behind the outside variation by ϕ hours. Further, the magnitude of variation inside the building is slightly less, by a factor of $[KB(K^2 + \omega^2)^{-1/2}]$, than the outside variation. The angular frequency of variation ω is $2\pi/24$ radians/hour ($= 0.262 \approx 1/4$). Typical values for the dimensionless ratio ω/K lie between $1/2$ and 1. For this range, the lag between inside and outside temperature is approximately 1.8 to 3 hours and the magnitude of the inside variation is between 89% and 71% of the variation outside. Fig.4.5.1 shows the 24-hour sinusoidal variation of the outside temperature for a typical moderate day as well as the temperature variation inside the building for a dimensionless ratio ω/K of unity, which corresponds to a time constant $1/K$ of approximately 4 hours. In sketching the latter curve, we have assumed that the exponential term has died out.

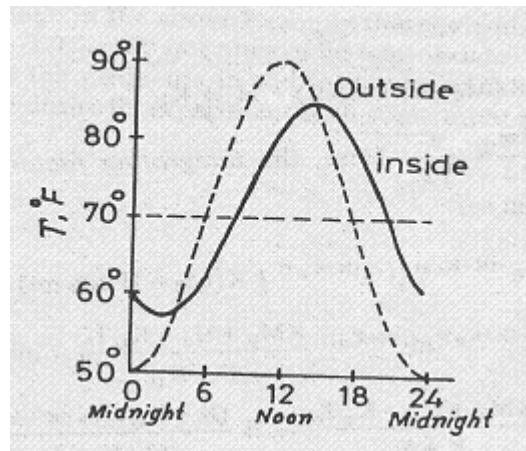


Fig. 4.5.1 Temperature variation inside and outside an unheated building

Example 4.5.3

Suppose that in Example 4.5.2, a simple thermostat is installed in the building which is used to compare the actual temperature inside the building with a desired temperature T_D . If the actual temperature is below the desired temperature, the furnace supplies heating; otherwise it is turned off. If the actual temperature is above the desired temperature, the air conditioner supplies cooling; otherwise it is turned off. (In practice there is some dead zone around the

desired temperature in which the temperature difference is not sufficient to activate the thermostat, but that is to be ignored here.) Assuming that the amount of heating or cooling supplied is proportional to the difference in temperature given as,

$$U(t) = K_U [T_D - T(t)],$$

where, K_U is the (positive) constant of proportionality. Find $T(t)$.

Solution

Substituting the value of $U(t)$ in the differential equation (4.5.1), we obtain the rate of change of the building-temperature as,

$$\frac{dT}{dt} = K [M(t) - T(t)] + H(t) + K_U [T_D - T(t)], \quad (4.5.7)$$

$$\text{i.e., } \frac{dT}{dt} + [K + K_U] T(t) = [K M(t) + H(t) + K_U T_D]. \quad (4.5.8)$$

It is a linear differential equation of the form $\frac{dT}{dt} + P(t) T = Q(t)$, so that,

$$P(t) = [K + K_U], \text{ and } Q(t) = [K M(t) + H(t) + K_U T_D].$$

If the additional heating rate $H = H_0$ (constant), the outside temperature $M(t)$ varies as a sinusoidal wave over a 24-hour period (like in the last example), then the *forcing function*, $Q(t)$, is given as,

$$Q(t) = K(M_0 - B \cos \omega t) + H_0 + K_U T_D,$$

where, $\omega = \frac{2\pi}{24} = \frac{\pi}{12}$. Here, the *integrating factor* is $e^{\int (K+K_U) dt} = e^{(K+K_U)t}$. Thus the solution is given as,

$$\begin{aligned} T(t) &= e^{-(K+K_U)t} \left[\int e^{(K+K_U)t} \{ K(M_0 - B \cos \omega t) + H_0 + K_U T_D \} dt + C \right] \\ &= e^{-(K+K_U)t} \left[e^{(K+K_U)t} \frac{KM_0 + H_0 + K_U T_D}{K + K_U} - e^{(K+K_U)t} \frac{(K + K_U) \cos \omega t + \omega \sin \omega t}{(K + K_U)^2 + \omega^2} + C \right] \\ \text{i.e., } T(t) &= \frac{KM_0 + H_0 + K_U T_D}{K + K_U} - B \frac{(K + K_U) \cos \omega t + \omega \sin \omega t}{(K + K_U)^2 + \omega^2} + C e^{-(K+K_U)t}. \end{aligned} \quad (4.5.9)$$

The constant of integration C is chosen so that $T(0) = T_0$ (the initial temperature).

An analysis of the result

In the example discussed above, the *time-constant* for Eqn.(4.5.8) is $\frac{1}{p} = \frac{1}{K+K_U}$. It is also referred to as the *time-constant for the building with heating and air-conditioning* facilities. Through practical experience the following estimations are considered reasonable:

- a. For a typical heating and cooling system, K_U is little less than 2;
- b. For a typical building, $\frac{1}{2} < K < \frac{1}{4}$;
- c. Thus the sum $(K + K_U) \approx 2$;
- d. Hence, the *time-constant* for a building with heating and cooling is about $\frac{1}{2}$ hr.

Therefore, when heating or cooling is turned ON, it takes about half an hour for the exponential term in Eqn.(4.5.9) to die down. If this exponential term is ignored, the average temperature inside the building is given by,

$$\frac{KM_0 + H_0 + K_U T_D}{K + K_U}.$$

Here, the term $(K + K_U)$ is much larger than K , whereas H_0 is small. Therefore, the above quantity reduces to, approximately, T_D , the *desired temperature*. This implies that after some time the temperature inside the building is approximately T_D , with a small sinusoidal variation. In fact the outside temperature M_0 and the inside heating rate H_0 , have only a negligible effect. Therefore, in order to save energy it is advisable to keep the heating or cooling system switched OFF at night. The same may be switched ON next morning and it will take about half an hour for the inside of the building to attain the desired temperature.

Note.

The assumption made in the above example that the amount of heating or cooling is give as $U(t) = K_U[T_D - T(t)]$ may not always be appropriate. This assumption has been made here as well as in the problems in the following exercise to illustrate the use of *time-constant*. One may desire to experiment with some other model for $U(t)$.

EXERCISE 4.2

[Newton's Law of Cooling – Variables Separable Form]

1. A thermometer which reads 75° F indoor is taken outdoor. After 5 minutes it reads 65° ; after another 5 minutes it reads 60° . What is the outdoor temperature? **[55°]**

2. Mr. Hassan and Mr. Mohsin both order coffee at the lunch counter and receive their cups at the same time. Mr. Hassan adds cream at once but does not take his coffee until 5 minutes later. Mr. Mohsin waits for 5 minutes, then adds cream and begins to drink. Who takes the hotter coffee? **[Mr. Mohsin]**

3. A thermometer, reading 10° is brought into a room having a temperature of 18° . One minute later the thermometer reads the temperature as 14° . How long will it take until the reading is practically 18° , say 17.9° ? **[About $6 \frac{1}{2}$ minutes]**

4. From a room, a thermometer is taken outside where the temperature is 5° F. After 1 minute the thermometer reads 55° F, and after 5 minutes it reads 30° F. Determine the initial temperature of the room. **[$T(0) = 64.4611^{\circ}$ F]**

5. The temperature of a room is 70° F. A thermometer is taken from that room to the outside where the temperature is 10° F. After 30 seconds the thermometer reads 50° F. Determine the reading at $t = 1$ minute. How long will it take for the thermometer to read 15° F? **[$T(1) = 36.67^{\circ}$ F; 3.06 min. approx]**

6. If the temperature of air is 20° , how long will it take an object to cool from 40° to 30° if it takes 10 minutes to reach 35° ? **[$10 \{ \ln 2 / \ln (4/3) \} = 24.1$ minutes]**

7. What will be the temperature of a body at the end of 10 minutes, if it cools from 35° to 30° in 3 minutes in air at 25° ? **[$25 + 10 \cdot 2^{-10/3} = 26^{\circ}$]**

8. What was the initial temperature of a body if the readings at 4 and 7 minutes were 50° and 45° , respectively, in a 40° medium? **[65.2°]**

9. **Stefan's Law of Cooling** (useful for greater temperature ranges than Newton's Law) states that the temperature of a body decreases at a rate proportional to the difference of the fourth powers of the temperature of the body and that of the surrounding medium. If the

temperature of a body decreases from 100° to 40° in 10 minutes, in air at 20° , derive a formula for the time t to reach at temperature T .

$$[\mathbf{0.0161} \, t = \ln \frac{T+20}{T-20} + 2 \tan^{-1} \frac{T}{20} - \mathbf{3.151}]$$

10. A body in a room at 60° cools from 200° to 120° in half an hour.

- Show that its temperature after t minutes is $(60+140 e^{-kt})$, where $k = (\ln 7 - \ln 3)/30$.
- Show that the time t required to reach a temperature of T degrees is given by the formula:
 $t = [\ln 140 - \ln (T-60)]/k$, where $60 < T \leq 200$.
- Find the time at which the temperature is 90° .
- Find a formula for the temperature of the body at time t if the room temperature is not kept constant but falls at a rate of 1° each 10 minutes. Assume the room temperature to be 60° when the body temperature is 200°

$$[\mathbf{c. 54.5 \text{ minutes}; d. } T = \frac{1}{10k} \{1 + (600 - t)k + (1400k - 1)e^{-kt}\}]$$

11. A thermometer has been stored in a room whose temperature is 75° . Five minutes after being taken outdoors it reads 65° . After another 5 minutes it reads 60° . Compute the outdoor temperature.
[55°]

12. On a Friday morning while people are working inside an office, the air conditioner keeps the temperature inside the building at 24° C. At noon the air conditioner is turned off and the people go home. The temperature outside is a constant 35° C the rest of the afternoon. If the time constant for the building is 4 hours, what will be the temperature inside the building at 2 pm? At 6 pm? When will the temperature inside the building reach 27° C?
[28.3° C; 32.5° C; 1:16 PM]

4.6 Dilution and Mixture of Liquids

The problems related to mixing of fluids lead to first order linear differential equations. In this section we shall discuss such problems.

Example 4.6.1

A tank contains 100 gallons of brine whose concentration is 2.5 pounds of salt per gallon. Brine containing 2 pounds of salt per gallon runs into the tank at a rate of 5 gallons per minute and the mixture (kept uniform by stirring) runs out at the same rate. Find the amount of salt in the tank at every instant.

Solution

Let $y(t)$ denote the number of pounds of salt in the tank at time t minutes after mixing begins. There are two factors which cause y to change:

- The *incoming brine* which brings salt in at a rate of 10 pounds per minute, and
- The *outgoing mixture* which removes salt at a rate of $5(y/100) = y/20$ pounds per minute (where the fraction $y/100$ represents the concentration at time t).

For the problems of the type as given above, the net rate at which $y(t)$ changes is given by,

$$\frac{dy}{dt} = (\text{rate of the substance entering}) - (\text{rate of the substance leaving}).$$

Therefore, the given problem can be expressed mathematically by the following differential equation,

$$dy/dt = 10 - y/20 = (200 - y)/20, \quad y(0) = 250.$$

It is a linear differential equation of first order, of “separation of variables” form. On integration it gives,

$$\ln |200 - y| = -t/20 + \ln c = \ln ce^{-t/20}.$$

It gives,

$$200 - y = ce^{-t/20},$$

or,

$$y(t) = 200 - ce^{-t/20}.$$

Using the initial condition this equation gives $c = -50$. Thus we get,

$$y(t) = 200 + 50 e^{-t/20}.$$

This equation gives us the amount of salt at any instant t .

The above solution shows that $y > 200$ for all t and that $y \rightarrow 200$ as t increases without bound (i.e., as $t \rightarrow \infty$). Hence the minimum salt content is 200 pounds. (This is also evident from the statement of the question).

The above equation can also be solved for t to yield,

$$t = 20 \ln \left[\frac{50}{y - 200} \right].$$

This enables us to find the time at which the salt content will be a given amount y , provided, $200 < y < 250$.

Example 4.6.2

A tank contains 10 gallons of fresh water, and a concentrated brine having 3 pounds of salt per gallon, is flowing into it at the rate of 2 gallons per minute. The solution is kept uniform by stirring, and it flows out at the same rate. How much salt is left in the tank at the end of 5 minutes?

Solution

Let $y(t)$ = Number of pounds of salt at the end of t minutes. Then, for the problems of the type as given above, the net rate at which $y(t)$ changes is given by,

$$\frac{dy}{dt} = (\text{rate of the substance entering}) - (\text{rate of the substance leaving})$$

But dy/dt = Number of pounds of salt entering the tank per minute (rate of increase of y).

Since 2 gallons of brine is flowing into the tank having concentration of 3 pounds per gallon, so ,

The number of pounds of salt entering the tank = 6 pounds per minute, and,

The number of pounds of salt leaving the tank = $2(y/10) = y/5$ pounds per minute,
(since each gallon in the tank contains $(y/10)$ pounds of salt).

Therefore, the given question can be expressed mathematically as,

$$\frac{dy}{dt} = 6 - \frac{2y}{10} = \frac{30 - y}{5}, y(0) = 0.$$

On integration it gives,

$$\ln |30 - y| = -t/5 + \ln c = \ln ce^{-t/5}.$$

It gives,

$$y(t) = 30 - ce^{-t/5}$$

Using the initial condition it gives $c = 30$. Thus the above equation becomes,

$$y(t) = 30 - 30 e^{-t/5}.$$

At the end of 5 minutes (taking $t = 5$), it gives the quantity of salt in the tank as,

$$y(5) = 30 - 30 e^{-1} = 19 \text{ pounds (approx.)}.$$

Example 4.6.3

Brine, containing 1 pound of salt per gallon, runs into a tank initially filled with 100 gallons of fresh water at the rate of 5 gallons per minute. The mixture runs out at the same rate. Find the amount of salt in the tank at the end of 100 minutes, assuming that the mixture is kept uniform by stirring.

Solution

Let $y(t)$ = the quantity of salt in the tank (of 100 gallons) at time t . So,

$y/100$ = the quantity of salt in 1 gallon.

Then, for the problems of the type as given above, the net rate at which $y(t)$ changes is given by,

$$\frac{dy}{dt} = (\text{rate of the substance entering}) - (\text{rate of the substance leaving})$$

Now, the number of pounds of salt entering the tank in 1 minute = 5, and,

The number of pounds of salt leaving the tank in 1 minute, @ 5 gallons/minute = $5(y/100)$,
(since each gallon in the tank contains $(y/100)$ pounds of salt).

Now, the rate of change of y w.r.t time t is given as,

The rate at which the salt runs in – the rate at which it runs out.

This yields the differential equation,

$$\frac{dy}{dt} = 5 - 5\frac{y}{100} = \frac{100 - y}{20}, \text{ i.e., } \frac{dy}{100 - y} = \frac{dt}{20}.$$

On integration it gives,

$$\ln |100 - y| = -t/20 + \ln c = \ln ce^{-t/20}.$$

Thus we get,

$$y(t) = 100 - ce^{-t/20}.$$

For the given initial condition it gives $c = 100$. Thus the above equation becomes,

$$y(t) = 100 - 100 e^{-t/20} = 100 (1 - e^{-t/20}).$$

Therefore, the amount of salt in the tank at the end of 100 minutes is given by,

$$y(100) = 100 (1 - e^{-5}) = 99.33 \text{ pounds.}$$

Example 4.6.4

Initially 50 pounds of salt is dissolved in a large tank having 300 gallons of water. A brine solution is let into the tank at the rate of 3 gallons per minute. The solution is kept homogeneous by stirring and is pumped out of the tank at the same rate. If the concentration of the solution entering the tank is 2 pounds per gallon, calculate the amount of salt in the tank at any time t . How much salt is present after 50 minutes? After a long time?

Solution

Let $y(t)$ = number of pounds of salt in the tank at any time t .

For the problems of the type as given above, the net rate at which $y(t)$ changes is given by,

$$\frac{dy}{dt} = (\text{rate of the substance entering}) - (\text{rate of the substance leaving}) = x_1 - x_2.$$

In the given problem, the rate at which salt enters the tank is given as,

$$x_1 = (3 \text{ gallons/minute}) \cdot (2 \text{ pounds/gallon}) = 6 \text{ pounds/minute, and}$$

the rate at which the salt is leaving the tank is given as,

$$x_2 = (3 \text{ gallons/minute}) \cdot (y/300 \text{ pounds/gallon}) = y/100 \text{ pounds/minute.}$$

Thus we have the differential equation as,

$$\frac{dy}{dt} = 6 - \frac{y}{100},$$

$$\text{i.e.,} \quad \frac{dy}{600 - y} = \frac{dt}{100}, \quad y(0) = 50.$$

Integrating both sides this equation gives,

$$\ln |600 - y| = -t/100 + \ln c = \ln ce^{-t/100}.$$

It gives,

$$y(t) = 600 - ce^{-t/100}.$$

Using the given initial condition it gives $c = 550$. Thus the above equation becomes,

$$y(t) = 600 - 550 e^{-t/100},$$

which gives the amount of salt in the tank at any time t .

To find the amount of salt in the tank after 50 minutes, we put $t = 50$ in the above equation to get,

$$y(50) = 600 - 550 e^{-1/2} = 266.408 \text{ pounds.}$$

To find the amount of salt left in the tank after a long time we let $t \rightarrow \infty$ in the above equation. Thus we get,

$$y(\infty) = 600 - 0 = 600 \text{ pounds.}$$

Example 4.6.5 (Rate of Pumping in the Solution > the Rate of Pumping out)

Initially 50 pounds of salt is dissolved in a large tank having 300 gallons of water. A brine solution is let into the tank at the rate of 3 gallons per minute. The solution is kept homogeneous by stirring and is pumped out of the tank at the rate of 2 gallons per minute. If the concentration of the solution entering the tank is 2 pounds per gallon, calculate the amount of salt in the tank at any time t .

Solution

Let $y(t)$ = number of pounds of salt in the tank at any time t . For the problems of the type as given above, the net rate at which $y(t)$ changes is given by,

$$\frac{dy}{dt} = (\text{rate of the substance entering}) - (\text{rate of the substance leaving}) = x_1 - x_2.$$

In the given problem, the rate at which salt enters the tank is given as,

$$x_1 = \left(3 \frac{\text{gallons}}{\text{minute}}\right) \cdot \left(2 \frac{\text{pounds}}{\text{gallon}}\right) = 6 \frac{\text{pounds}}{\text{minute}},$$

and, in this case the solution is accumulating at a rate of $(3-2)$ gallons/minute = 1
gallon/minute.

After t minutes there are $(300+t)$ gallons of brine in the tank. So, the rate at which the salt is leaving the tank is given as,

$$x_2 = (2 \text{ gallons/minute}) \cdot \left(\frac{y}{300+t} \text{ pounds/gallon}\right) = \frac{2y}{300+t} \text{ pounds per minute.}$$

Thus the given problem reduces to,

$$\frac{dy}{dt} = x_1 - x_2 = 6 - \frac{2y}{300 + t}, \text{ i.e., } \frac{dy}{dt} + \frac{2}{300 + t} y = 6, y(0) = 50.$$

It is of the form $\frac{dy}{dt} + P(t) y = Q(t)$, wherein the integrating factor is, $e^{\int P dt} = (300 + t)^2$. Thus

the solution of the above equation is given as, $y(t) = e^{-\int P dt} \left[\int e^{\int P dt} Q(t) dt + c \right]$, i.e.,

$$y(t) = (300 + t)^{-2} \left[\int (300 + t)^2 \cdot 6 dt + c \right] = (300 + t)^{-2} [6 + c].$$

Thus,

$$y(t) = 2 (300 + t) + c (300 + t)^{-2}.$$

Using the initial condition $y(0) = 50$ this equation gives $c = -4.95 \times 10^7$. Thus the above equation becomes,

$$y(t) = 2 (300 + t) - (4.95 \times 10^7) (300 + t)^{-2},$$

which gives the amount of salt in the solution at any time t .

Note. The above solution is valid for $0 \leq t \leq T$, where T is the time at which the tank overflows.

EXERCISE-4.3

(Dilution or Mixture of Liquids – Separation of Variables Form)

1. A tank containing 300 gallons of saturated salt solution (3 pounds of salt per gallon) is to be diluted by the addition of brine containing 1 pound of salt per gallon. If this solution runs into the tank at a rate of 4 gallons per minute and the mixture runs out at the same rate, when will the concentration in the tank reach 1.03 pounds per gallon?

$$[t = 75(\ln 600 - \ln 9) \approx 5 \text{ hours } 15 \text{ minutes}]$$

2. In a tank are 100 gallons of brine containing 60 pounds of dissolved salt. Pure water runs into the tank at the rate of 2 gallons per minute, the mixture being kept uniform by stirring. The mixture runs out at the rate of 3 gallons per minute. How much salt is in the tank at the end of $1 \frac{1}{2}$ hours?

$$[0.06 \text{ lbs}]$$

3. In a tank are 100 gallons of brine containing 50 pounds of dissolved salt. Water runs into the tank at the rate of 3 gallons per minute, and the concentration is kept uniform by stirring. How much salt is in the tank at the end of one hour if the mixture runs out at a rate of 2 gallons per minute?

$$[19.5 \text{ pounds}]$$

4. A tank contains 200 liters of fluid in which 30 g of salt is dissolved. Brine containing 1 g of salt per liter is then pumped into the tank at a rate of 4 liters per minute. The well-mixed solution is then let out at the same rate. Find the number of grams of salt $y(t)$ in the tank at any time t .

$$[y = 200 - 170 e^{-t/50}]$$

5. A large tank is filled with 500 gallons of pure water. Brine containing 2 pounds of salt per gallon is pumped into the tank at a rate of 5 gallons per minute. The solution, kept homogeneous by stirring, is pumped out at the same rate. Find the number of pounds of salt $y(t)$ in the tank at any time t .

$$[y(t) = 1000(1 - e^{-t/100})]$$

6. Solve problem 7 under the assumption that the solution is let out at a faster rate of 10 gallons per minute. When would the tank be empty?

$$[y(t) = 1000 - 10t + c(100 - t)^2; \text{ 100 minutes}]$$

7. A large tank is partially filled with 100 gallons of water in which 10 pounds of salt is dissolved. Brine containing $\frac{1}{2}$ pound of salt per gallon is pumped into the tank at a rate of 6 gallons per minute. The solution, kept uniform by stirring, is pumped out at a slower rate of 4 gallons per minute. Find the number of pounds of salt in the tank after half an hour.

$$[64.38 \text{ pounds}]$$

8. Into a tank originally containing 20 pounds of salt dissolved in 8 gallons of water, there flows fresh water at the rate of 3 gallons per minute, and the solution, constantly stirred, flows out at the same rate. How much salt is in the tank at the end of 4 minutes?

$$[20 e^{-3/2} = 4.46 \text{ pounds}]$$

9. Solve problem 10 assuming that the water flows into the tank at a rate of 4 gallons per minute and flows out at a rate of 2 gallons per minute.

$$[10 \text{ pounds}]$$

10. Two gallons of water per minute, containing 2 pounds of salt per gallon, flow into a tank initially containing 10 gallons of fresh water. If the solution flows out at the same rate, how much salt is in the tank at the end of 5 minutes?

$$[20(1 - 1/e) = 12.6 \text{ pounds}]$$

11. In problem 12 find the amount of salt at the end of $2\frac{1}{2}$ minutes assuming that there are originally 5 gallons of fresh water and that the inflow and outflow rates are, respectively, 3 gallons/minute and 2 gallons/minute.

$$[95/9 = 10.56 \text{ pounds}]$$

12. A saturated (3 pounds per gallon) salt solution flows into a tank containing 4 pounds of salt dissolved in 5 gallons of water, at a rate of 2 gallons per minute. If the solution flows out at the same rate, when will there be $9\frac{1}{2}$ pounds of salt in the tank?

$$[(5/2) \ln 2 = 1.73 \text{ minutes}]$$

13. In problem 14 when will there be 7 pounds of salt in the tank if the inflow and outflow rates are, respectively, 1gallon/minute and 2 gallons/minute, and the tank originally contains 10 pounds of salt dissolved in 6 gallons of water?

$$[3 \text{ minutes}]$$

14. Brine containing 3 pounds of salt per gallon flows into a tank at a rate of 4 gallons per minute. If the stirred solution flows out at the rate of 6 gallons per minute, when will the concentration be 1.4 pounds per gallon? The tank originally contains 10 pounds of salt in 20 gallons of water.

$$[2 \text{ mins}]$$

15. In problem 16 when will the concentration be 2 pounds per gallon if there are initially 10 gallons of solution in the tank and the inflow and outflow rates are both 1 gallon per minute?

$$[10 \ln 2 = 6.93 \text{ minutes}]$$

16. A tank initially contains 10 pounds of salt dissolved in 5 gallons of water. What is the concentration of salt in a solution entering at a rate of 4 gallons per minute if the stirred solution leaves at the same rate and there are 5 pounds of salt in the tank at the end of $2\frac{1}{2}$ minutes?

$$[\frac{e^2 - 2}{e^2 - 1} = 0.84 \text{ pounds/gallon}]$$

17. A certain quantity of salt in a container is dissolved in 10 gallons of water. A brine containing 2 pounds of salt per gallon is allowed to enter the container at a rate of 5 gallons

per minute and the stirred solution to leave at the same rate. If there are 16 pounds of salt in the container at the end of 2 minutes, what was the initial quantity of salt?

$$\mathbf{[4 (5 - e) = 9.1 \text{ pounds }]}$$

4.7 Flow of Fluid Through an Orifice – Torricelli's Law

Consider a water tank (not necessarily cylindrical) having a hole at its bottom, from which water is flowing out. Under the ideal conditions, if there were no friction (and hence no loss of energy), the **Torricelli's Law** states that,

The velocity v , of water exiting through the hole, would be given by,

$$v = \sqrt{2gh(t)},$$

where, $g = 9.8 \text{ m/sec}^2 = 32 \text{ ft/sec}^2$ is the acceleration due to gravity of the earth, and $h(t)$ is the height of water above the hole at time t .

The above equation gives the velocity of a drop of water falling freely from a certain quantity of water through a hole (or orifice). However, under the real conditions, because of friction the jet-stream contracts somewhat and the actual rate of discharge of water is given by,

$$\frac{dh(t)}{dt} = -c \sqrt{2gh(t)}, \quad (4.7.1)$$

where c is an experimentally determined number, called **discharge coefficient**, having value between 0 and 1. For ordinary small continuous stream of water through sharp-edged orifice, the approximate value of c is 0.6. It was introduced by **J. C. Borda** in 1766 because the stream flowing out has a smaller cross-sectional area than the area of the hole in the tank. The negative sign in the above equation indicates that the height of water h decreases with time t . Thus equation (4.7.1) becomes,

$$\begin{aligned} \frac{dh(t)}{dt} &= -0.6 \sqrt{2gh(t)} \\ &= -4.8 \sqrt{h(t)} \text{ ft/sec} \end{aligned} \quad (4.7.2)$$

$$= -26.56 \sqrt{h(t)} \text{ cm/sec.} \quad (4.7.3)$$

Let A_0 = cross-sectional area of the orifice. On multiplying A_0 on both sides of the above equation we get,

$$\begin{aligned} \frac{dh(t)}{dt} A_0 &= -4.8 A_0 \sqrt{h(t)} \text{ ft}^3/\text{sec.} \\ &= \text{Rate of discharge of volume, flowing through the orifice} = \frac{dV(h)}{dt}. \end{aligned}$$

Thus,

$$\frac{dV(h)}{dt} = -4.8 A_0 \sqrt{h(t)} \text{ ft}^3/\text{sec.} \quad (4.7.4)$$

Letting,

$A(u)$ = cross-sectional area of the tank at height u ,

$V(h)$ = Volume of the fluid in the tank, when height of fluid in it is h ,

we can write,

$$V(h) = \int_0^h A(u) du. \quad (4.7.5)$$

Differentiating this equation w.r.t. h we obtain,

$$\frac{dV(h)}{dh} = A(u)]_0^h = A(h) - A(0).$$

But $A(0) = 0$. Thus we get,

$$\frac{dV(h)}{dh} = A(h). \quad (4.7.6)$$

But by chain rule, $\frac{dV}{dt} = \frac{dV}{dh} \frac{dh}{dt}$. Therefore, substituting the values from Eqns.(4.7.4) and (4.7.6) we obtain,

$$-4.8 A_0 \sqrt{h(t)} = A(h) \frac{dh}{dt}.$$

It gives,

$$\frac{dh}{dt} = -4.8 \frac{A_0}{A} \sqrt{h(t)} \text{ ft/sec} \quad (\text{British System}) \quad (4.7.7)$$

$$= -26.56 \frac{A_0}{A} \sqrt{h(t)} \text{ cm/sec.} \quad (\text{Metric System}) \quad (4.7.8)$$

Therefore, if a tank filled with water is allowed to discharge the water through an orifice under the influence of gravity, the above equations give the **rate of change of the height h of water in the tank at any time t** .

Example 8.7.1

A cylindrical tank 1.50 meters high stands on its circular base of diameter 1.00 meter and is initially filled with water. At the bottom of the tank there is a hole of diameter 1.00 cm, which is opened at some instant, so that the water starts draining out under the influence of gravity. Find the height $h(t)$ of the water in the tank at any time t . Also find the time at which the tank is,

a. One-half full,

- b. One-quarter full, and,
- c. Empty.

Solution

Here, cross sectional area of the tank is: $A(h) = \pi (50)^2 \text{ cm}^2$, and
cross sectional area of orifice (hole) is: $A_0 = \pi(0.5)^2 \text{ cm}^2$.

Thus $A_0/A(h) = 0.0001$, so that Eqn. (4.7.8) gives,

$$\frac{dh}{dt} = -0.002656\sqrt{h(t)}.$$

On integration it yields,

$$h(t) = (c - 0.001328 t)^2. \quad (4.7.9)$$

Now, at $t = 0$, the height of the tank is: $h(0) = 150 \text{ cm}$ (initial condition). So the above equation gives, $c = 12.25$. Thus Eqn.(4.7.9) becomes,

$$h(t) = (12.25 - 0.001328 t)^2, \quad (4.7.10)$$

which gives the height of water at any time t . Solving this equation for t we get,

$$t = \frac{12.25 - \sqrt{h}}{0.001328} = 9224.4 - 753 \sqrt{h}. \quad (4.7.11)$$

- a. To find the time after which the tank would be half full we put $h = 75 \text{ cm}$ in the above equation to get,

$$t = 2793 \text{ secs.} = 45 \text{ mins. approx.}$$

- b. To find the time when the tank would be quarter full we put $h = 37.5 \text{ cm.}$ in Eqn.(4.7.11) to get,

$$t = 4613 \text{ secs.} = 77 \text{ mins. approx.}$$

- c. To find the time when the tank would be empty we put $h = 0$ in the above equation to get,

$$t = 9224.4 \text{ secs.} = 154 \text{ mins. approx} = 2 \text{ hours } 34 \text{ mins. approx}$$

Example 4.7.2

Consider a tank with constant area $A(h) = A$, say. If the level of water is lowered from 10 feet to 9 feet in 10 minutes (600 secs.), find the time taken to lower the water level from 7 feet to 1 foot.

Solution

In this case we are given: $h(0) = 10$ feet, and $h(600) = 9$ feet. Thus Eqn. (4.7.7) gives,

$$\int_{10}^9 \frac{dh}{\sqrt{h}} = - \int_0^{600} k dt, \text{ where } k = -4.8 \frac{A_0}{A}$$

Integrating this equation we get $k = 5.41 \times 10^{-4}$.

To find the time taken by the water to lower from 7 feet to 1 foot we have,

$$\int_7^1 \frac{dh}{\sqrt{h}} = -5.41 \times 10^{-4} \int_{t_1}^{t_2} dt.$$

On integration it gives,

$$t_2 - t_1 = 101.4 \text{ minutes,}$$

which is the required time taken by the water to lower from 7 feet to 1 foot in the given tank.

EXERCISE – 4.4

[Torricelli's Law – Variables Separable Form]

Note: Answers are given against the respective questions within square brackets.

1. A tank with vertical sides has a square cross-section of area 4 square feet. Water is leaving the tank through an orifice of area $5/3$ square inches. If the water level is initially 2 feet above the orifice, find the time required for the level to drop 1 foot.

[59.6 secs.]
2. Refer to the preceding problem. If water also flows into the tank at a rate of 100 cubic inches per second, show that the water level approaches the value $(25/24)^2$ feet above the orifice, regardless of the initial water level.
3. A tank has a shape of a right circular cone with its vertex up. Find the time required to empty a liquid from the tank through an orifice in its base. Express your result in terms of the dimensions of the cone and the area A_0 of the orifice.

$[2\pi R^2 \sqrt{h} / 9A_0 \text{ secs, where } R \text{ is the radius of the base and } h \text{ the height of the cone (in feet)}]$
4. In Example 4.6.2 it is claimed that the speed a body gains in free fall from rest through height h (without air resistance) is $\sqrt{2gh}$. Derive it.
5. If the diameter of the hole in Example 4.6.2 is doubled and all other data are left as before, how will the answer change?

[11.3, 19.2, 38.4]
6. Suppose that the tank in example 4.6.2 is hemispherical of radius R , which is initially full of water, having an orifice of 5 cm^2 cross-sectional area at the bottom. The outlet is opened at some instant. Find the time it takes to empty the tank (i) for any given R ; (ii) for $R = 1$ meter.

[(i) $t = 0.0221 R^{5/2}$; (ii) $t = 37 \text{ mins.}$]
7. Find the time to empty a cylindrical tank 2 feet in diameter and 3 feet high through a hole 2 inches in diameter in the bottom of the tank. Initially the tank is full of water and its axis is (i) vertical; (ii) horizontal. **[(i) 1 min. 44 secs.; (ii) 1 min. 48 secs.]**
8. Use Eqn. (4.7.6), properly modified, to find the time required to fill a cubical tank of edge 3 feet if there is a round hole 1 inch in diameter in the bottom of the tank and if water is poured into the tank at $\pi \text{ feet}^3/\text{minute}$.

[26 mins. 28 secs.]
9. The rate at which water runs out of a tank through a small hole is proportional to the square root of the height of the surface of the water above the hole. If the tank is a right circular cone with vertex down and axis vertical and if water runs out through a small hole at the bottom

and if the surface of the water is originally 20 inches above the vertex and falls 9 inches in 60 minutes, find the time required to empty the tank.

$$[t = \frac{120\sqrt{5}}{2\sqrt{5} - \sqrt{11}} = 232.22 \text{ mins.} = 3 \text{ hrs. } 52 \text{ mins.}]$$

4.8 Compound Interest

Let $P(t)$ be the amount at time t , on which a bank pays a yearly interest at the rate of r %, compounded continuously. Then,

$$\frac{dP}{dt} = \frac{r}{100} P \quad (t \text{ in years}). \quad (4.8.1)$$

It is a first order linear differential equation of *variables separable* form.

On integrating it gives,

$$P(t) = c e^{(r/100)t}.$$

If $P(0) = P_0$, the initial amount at $t = 0$, or the principal amount, then the above equation gives: $c = P_0$. Therefore, this equation becomes,

$$P(t) = P_0 e^{(r/100)t}, \quad (4.8.2)$$

which gives the *compound interest* at the *rate of* r % on the *principal amount* P_0 , after the lapse of t years.

Example 4.8.1

Suppose the interest rate is 5%, the principal amount $P_0 = \text{Rs } 1000$, and no amount is withdrawn,

- How much amount will be in the account after 2 years?
- When will the account reach Rs 4000?

- (c) If Rs 1000 are added to the account after 12 months, how much will be in the account after $3\frac{1}{2}$ years?

Solution

- (a) Taking $P_0 = 1000$, $r = 5$, and $t = 2$, Eqn. (4.8.2) gives,

$$P(2) = 1000 e^{(0.05)^2} = \text{Rs } 1105.17.$$

- (b) Taking $P_0 = 1000$, $P(t) = 4000$, and $r = 5$, Eqn (4.8.2) gives: $4000 = 1000 e^{0.05 t}$.
Taking log-natural of both sides it gives: $0.05 t = \ln 4$. Thus we get,

$$t = \ln 4 / 0.05 = 27.73 \text{ years approx.,}$$

Thus Rs 1000 will grow to Rs 4000 in a period of 27.73 years approx.

- (c) Taking $P_0 = 1000$, $r = 5$, $t = 1$ year, we get,

$$P(1) = 1000 e^{(0.05)^1} = 1051.27.$$

Since an amount of Rs.1000 is added at the end of one year, so at that time amount would be, $P(t) = 1000 + 1051.27 = 2051.27$.

Next, taking $P(0) = 2051.27$, $t = 2\frac{1}{2}$, $r = 5$, Eqn (4.8.2) gives,

$$P(3\frac{1}{2}) = 2051.27 e^{(0.05)^{(2.5)}} = \text{Rs } 2324.40,$$

i.e., the amount in the account after $3\frac{1}{2}$ years would be Rs 2324.40

EXERCISE - 4.5

(Compound Interest)

Note: Answers are given against the respective questions within square brackets.

1. An investment of Rs 200,000 is made which earns interest at the rate of 8 % per year. If interest is compounded continuously:

- (a) Determine the exponential function which states the compound amount as a function of years of investment t .

(b) What will Rs 200,000 grow to if invested for 5 years? 10 years?

[(a) $200,000 e^{0.08t}$, (b) Rs 298,364.94, Rs 445,108.19]

2. In problem 1, determine the length of time required for the investment to double in value. To quadruple.
3. An investment of Rs 1 million which earns interest at the rate of 8.8 % per year is made. If interest is compounded continuously:

(a) What will the investment grow to if it is invested for 10 years? 25 years?

[38.66 years]

(b) How long will it take for the investment to increase by 50 percent?

[(a) Rs 2,339,646.80; Rs 8,372,897.50, (b) 4.77 years]

4. A certain sum of money P draws interest compounded continuously. If at a certain time the account has an amount of Rs P_0 , determine the time when the principal attains the value $2P_0$ rupees if the annual interest rate is (a) 2%; (b) 4%.

[(a) $50 \ln 2$ years; (b) $25 \ln 2$ years]

5. A bank offers 5% interest compounded continuously in a savings account. Determine (a) the amount of interest earned in 1 year on a deposit of Rs 100, and (b) the equivalent rate if the compounding were done annually.

[(a) Rs 5.13; (b) 5.13 %]

4.9 Electric Circuits and Networks

In this section we shall set up mathematical models for electric circuits leading to linear differential equations of first order. They would be of interest to the students of electrical and computer engineering in particular, and others in general. Here we shall discuss some basic concepts and definitions related to electric models.

- a. **Electromotive Force (EMF) or Voltage**. It is the force (pressure) which makes or tends to make electricity flow round a circuit. In case of a battery the difference of potentials between its two terminals is called its EMF or voltage. It is measured in joules/coulombs, or volts (1 joule/coulomb = 1 volt).
- b. **Resistor (or Electrical Resistor)**. It is a device (e.g., a piece of wire) designed to have a definite amount of resistance, used in circuits to limit current flow or to provide a voltage drop. Resistance is measured in ohms by the Ohm's Law:

$$\text{Resistance (in ohms)} = \frac{\text{pressure (in volts)}}{\text{current (in amperes)}}$$
- c. **Inductance**. When a changing current (e.g., alternating current) flows through a coil, pressure of electricity (voltage) is induced (set up, caused) in the coil. This pressure tries to stop the current from changing; this is *self induction*. The *self inductance* of a coil is a measure of its power to produce such a voltage. The unit of inductance is *henry*. A coil has an inductance of 1 henry if one volt is produced in it when the current through it changes at the rate of one ampere per second.
- d. **Inductor**. It is a piece of apparatus (e.g., a coil or choke etc) used in an electric circuit because it has inductance.
- e. **Capacitor or Condenser (electrical)**. Two nearby conductors (having any shape; plates etc) carrying equal and opposite charges, separated by air or some other insulator, form a capacitor. This arrangement of conductors can hold a large amount of electricity without pressure (voltage) between the conductors being high.
- f. **Capacitance**. It is the measure of the power of a capacitor. It is measured in *farads* and is the ratio of the charge on one of the conductors of the capacitor to the potential difference between the conductors.

We shall use the following notations which are common to most texts in electrical engineering:

t (seconds) = time;

Q (coulombs) = quantity of electricity; e.g., charge on a capacitor;

I (amperes) = current, time rate of flow of electricity ($\Rightarrow I(t) = \frac{dQ(t)}{dt}$)

E (volts) = electromotive force or voltage;

R (ohms) = resistance;

L (henrys) = inductance;
 C (farads) = capacitance.

The elements in a circuit diagram are denoted by conventional symbols as shown in Fig. 4.9.1, below:

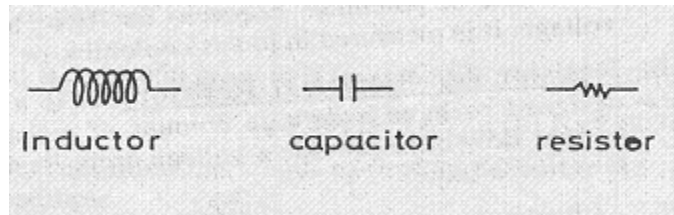


Fig. 4.9.1: Inductor, Capacitor and Resistor

The physical principles governing electric circuits were formulated by the German physicist **Gustav Robert Kirchhoff** (1824-1887) in 1859. They are as follows:

- Kirchoff's Current Law.** The algebraic sum of currents flowing into any junction point must be zero. Or, at any point of a circuit, the sum of inflowing currents equals the sum of the outflowing currents.
- Kirchoff's Voltage Law.** The algebraic sum of all voltage drops around a closed circuit is zero. Or, in any closed loop, the sum of all voltage drops equals the impressed EMF.

Kirchoff's current law implies that the same current, I , passes through each element in the series circuit..

In order to apply Kirchoff's Voltage Law we need to know the voltage drop across each element of the circuit. These voltage formulas are stated below (for details some introductory book on physics may be consulted):

- An electromotive force (EMF) E , e.g., due to a battery, drives an electric charge Q and produces a current I . The current and charge are related by:

$$I(t) = \frac{dQ}{dt}.$$

- According to Ohm's law, the voltage drop E_R across a resistor is proportional to the current I passing through the resistor. It implies that:

$$E_R = RI,$$

where the constant of proportionality R is called the **resistance**.

- c. It can be shown by **Faraday's law** and **Lenz's law** that the voltage drop E_L across an inductor is proportional to the instantaneous rate of change of the current I :

$$E_L = L \frac{dI}{dt},$$

where the constant of proportionality L is called the **inductance**.

- d. The voltage drop E_C across a capacitor is proportional to the electric charge Q on the capacitor:

$$E_C = \frac{1}{C} Q,$$

where the constant of proportionality C is called **capacitance**.

Now we consider simple electric circuits, the RL and RC series circuits and application of differential equations to them. These circuits consist of voltage or energy sources (e.g., batteries or generators), resistors, inductors and capacitors. These circuits are illustrated in Fig. 4.9.2, as shown below:



(i) The LR series circuit

(ii) The RC series circuit

Fig. 4.9.2

As quoted above the quantities R (resistance), L (inductance) and C (capacitance) in the above mentioned circuits are constants, whereas E (EMF or voltage) may be a constant or function of time. Using the Kirchhoff's Voltage Law (KVL) we can derive linear differential equations for the above quoted RL and RC series circuits, as under:

- a. **LR Series Circuit.** In a series circuit containing an inductor and a resistor, by Kirchhoff's Voltage Law, the sum of the voltage drop across the inductor E_L ($= L dI/dt$) and the voltage drop across the resistor E_R ($= RI$) is equal to the impressed voltage, $E(t)$, on the circuit, i.e.,

$$E_L + E_R = E(t).$$

Thus we get the linear differential equation for the current $I(t)$ as,

$$L \frac{dI}{dt} + RI = E(t), \quad (4.9.1)$$

where the inductance L and resistance R are constants. Sometimes the current $I(t)$ is called *response* of the system.

- b. **RC Series Circuit.** In a series circuit containing a capacitor and a resistor, by Kirchoff's Voltage Law, the sum of the voltage drop across the capacitor $E_C (= \frac{Q(t)}{C})$ and the voltage drop across the resistor $E_R (= RI)$ is equal to the impressed voltage, $E(t)$, on the circuit, i.e.,

$$E_R + E_C = E(t).$$

Thus for the series circuit shown in Fig. 4.9.2(ii), Kirchoff's Voltage Law (KVL) gives,

$$RI + \frac{Q(t)}{C} = E(t). \quad (4.9.2)$$

But we know that $I = dQ/dt$, so Eqn. (4.9.2) becomes the linear differential equation,

$$R \frac{dQ(t)}{dt} + \frac{Q(t)}{C} = E(t). \quad (4.9.3)$$

Example 4.9.1 (RL-Circuit)

When a switch is closed in a circuit containing a resistance R , an inductance L , and a battery which supplies a constant voltage E , the current I builds up at a rate defined by the Eqn. (4.9.1) as,

$$L \frac{dI}{dt} + RI = E.$$

Find the current as a function of time. How long will it take I to reach one-half its final value?

Solution.

The given differential equation can be written as,

$$\frac{dI}{dt} + \frac{R}{L} I = \frac{E}{L} \quad .$$

It is a linear differential equation of first order in I , which is of the form $y' + Py = Q$, with integrating factor as $e^{(R/L)t}$. Therefore, its solution is given as,

$$I(t) = e^{-(R/L)t} \left[\frac{L}{R} e^{(R/L)t} \frac{E}{L} + c \right] = \frac{E}{R} + c e^{-(R/L)t}.$$

Taking $I(0) = 0$, this equation gives: $c = -E/R$. Therefore, we get the solution as,

$$I(t) = \frac{E}{R} (1 - e^{-Rt/L}). \quad (4.9.4)$$

The exponential term will approach zero as t approaches infinity. Thus practically, after some time the current will be constant, equal to E/R . In fact, this is the value, the current would have had immediately (by Ohm's law), had there been no inductor in the circuit. It may be seen that this limit is independent of the initial value $I(0)$. Fig. 4.9.3 shows the particular solution given by Eqn.(4.9.4), with $I(0) = 0$:

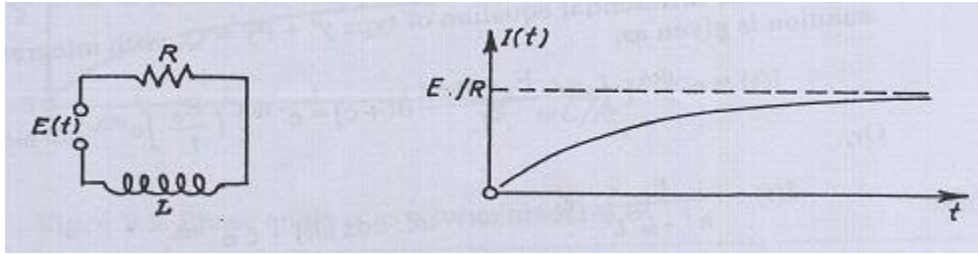


Fig. 4.9.3: Current in an RL-circuit due to a constant electromotive force = E/R

Next we calculate how long I will take to reach one-half its final value. The final value of I will be after a lapse of long time. For that we put $t = \infty$ in Eqn. (4.9.4) to get,

$$I_{\text{FINAL}} = \frac{E}{R}. \quad (4.9.5)$$

To find the requisite time we replace $I(t)$ in Eqn. (4.9.4) by $\frac{I_{\text{FINAL}}}{2} = \frac{E}{2R}$. Thus we get,

$$\frac{E}{2R} = \frac{E}{R} (1 - e^{-Rt/L}),$$

i.e.,

$$e^{-R t/L} = 0.5.$$

On solving it gives,

$$t = 0.693 \frac{L}{R}, \quad (4.9.6)$$

which is the time when the current I will be one-half its final value.

Example 4.9.2

In example 4.9.1, work out the current $I(t)$ at any time t if the electromotive force $E(t) = E_0 \sin \omega t$.

Solution

In this case we have to solve the differential equation,

$$L \frac{dI}{dt} + RI = E_0 \sin \omega t. \quad (4.9.7)$$

It can be written as,

$$\frac{dI}{dt} + \frac{R}{L} I = \frac{E_0}{L} \sin \omega t.$$

It is a linear differential equation of type $y' + Py = Q$, with integrating factor: $e^{tR/L}$. Its solution is given as,

$$I(t) = e^{-tR/L} \left[\int e^{tR/L} \frac{E_0 \sin \omega t}{L} dt + c \right] = e^{-tR/L} \left[\frac{E_0}{L} \int e^{tR/L} \sin \omega t dt + c \right].$$

Or,

$$I(t) = \frac{E_0}{R^2 + \omega^2 L^2} [R \sin \omega t - \omega L \cos \omega t] + c e^{-tR/L}. \quad (4.9.8)$$

Using the initial condition $I(0) = 0$, we get $c = \frac{E_0 \omega L}{R^2 + \omega^2 L^2}$. Therefore, Eqn. (4.9.8) gives the solution as,

$$I(t) = \frac{E_0}{R^2 + \omega^2 L^2} [\omega L e^{-tR/L} + R \sin \omega t - \omega L \cos \omega t].$$

To simplify this expression we multiply and divide it by $\sqrt{R^2 + \omega^2 L^2}$ and let

$$\cos \alpha = \frac{R}{\sqrt{R^2 + \omega^2 L^2}}, \text{ and } \sin \alpha = \frac{\omega L}{\sqrt{R^2 + \omega^2 L^2}} \text{ to get,}$$

$$I(t) = \frac{E_0}{\sqrt{R^2 + \omega^2 L^2}} [\omega L e^{-tR/L} + \sin(\omega t - \alpha)], \quad (4.9.9)$$

where, $\alpha = \tan^{-1}(\omega L/R)$. It may be seen that the exponential term in Eqn. (4.9.9) will approach zero as t approaches infinity. It means that after some time the current $I(t)$ would practically execute harmonic oscillations, as shown in Fig. 4.9.4, while Fig. 4.9.5 shows the phase angle α as a function of $\omega L/R$ (since $\alpha = \tan^{-1}(\omega L/R)$). Clearly, if $L = 0$ then $\alpha = 0$, and accordingly the oscillations of $I(t)$ are in phase with those of $E(t)$.

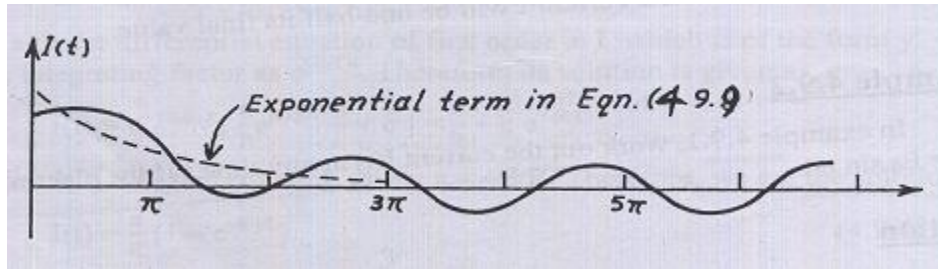


Fig. 4.9.4: Current in an RL circuit due to sinusoidal EMF as given by Eqn. (4.9.9)
(with phase angle $\alpha = \pi/4$)

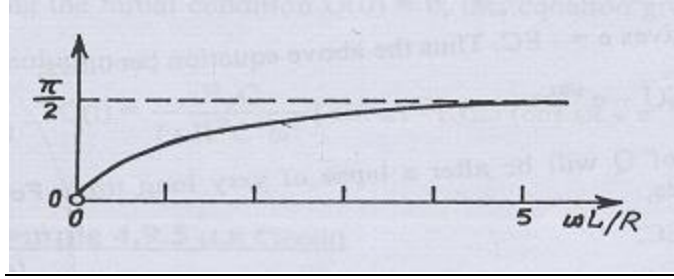


Fig. 4.9 5. Phase angle α as a function of $\omega L/R$

Steady State and Transient State

An electrical (or dynamical) system is said to be in the **steady state** when the variables describing its behaviour are periodic functions of time or constants. However, if such a system is not in the steady state, it is said to be in the **transient state** (or unsteady state). Accordingly the corresponding variables are called **steady state functions** and **transient functions** respectively. In example 4.9.1, the function E_0/R is a *steady state function* or *steady state solution* of Eqn. (4.9.1). In example 4.9.2 the steady state solution is represented by the last term in Eqn. (4.9.8). Before the circuit actually reaches the steady state it is in the transient state. In fact such an interim or transient period occurs because inductors and capacitors store energy, and the corresponding inductor currents and capacitor voltages cannot be changed instantly. Practically, this transient state will last for a short time.

Example 4.9.3 (RC Circuit)

When a capacitor of capacity C is being charged through a resistance R by a battery which supplies a constant voltage E , the instantaneous charge Q on the capacitor satisfies the differential equation,

$$R \frac{dQ}{dt} + \frac{Q}{C} = E. \quad (4.9.10)$$

Find Q as a function of time if the capacitor is initially uncharged, i.e., $Q_0 = Q(0) = 0$. How long will it be before the charge on the capacitor is one-half its final value?

Solution

The given differential equation can be written as,

$$\frac{dQ}{dt} + \frac{1}{RC} Q = \frac{E}{R},$$

which is a linear differential equation in Q with integrating factor as $e^{t/RC}$. Its solution is given as,

$$Q(t) = e^{-t/RC} \left[RC e^{t/RC} \frac{E}{R} + c \right] = EC + c e^{-t/RC}.$$

For $Q(0) = 0$ it gives $c = -EC$. Thus the above equation becomes,

$$Q(t) = EC(1 - e^{-t/RC}). \quad (4.9.11)$$

The final value of Q will be after a lapse of very long time. For that taking $t \rightarrow \infty$, Eqn. (4.9.11) gives,

$$Q_{\text{FINAL}} = EC. \quad (4.9.12)$$

To find t when Q will be one-half its final value we replace $Q(t)$ in Eqn. (4.9.11) by

$$\frac{Q_{\text{FINAL}}}{2} = \frac{EC}{2}. \text{ Thus Eqn. (4.9.11) gives,}$$

$$\frac{EC}{2} = EC(1 - e^{-t/RC}), \text{ i.e., } e^{-t/RC} = 0.5.$$

Taking log natural of both sides this equation and solving for t we get,

$$t = 0.693 RC, \quad (4.9.13)$$

which is the time when charge Q on the capacitor will be one-half its final value.

Example 4.9.4 (RC Circuit)

Work out $Q(t)$ in Example 4.8.3 if the battery is replaced by a generator which supplies an alternating voltage equal to $E_0 \sin \omega t$.

Solution

In this case we shall solve the following differential equation,

$$R \frac{dQ}{dt} + \frac{Q}{C} = E_0 \sin \omega t. \quad (4.9.14)$$

It can be written as,

$$\frac{dQ}{dt} + \frac{1}{RC}Q = \frac{E_0 \sin \omega t}{R}.$$

It is also a linear differential equation in $Q(t)$ with an integrating factor $e^{t/RC}$. Thus its solution is,

$$\begin{aligned} Q(t) &= e^{-t/RC} \left[\frac{E_0}{R} \int e^{t/RC} \sin \omega t \, dt + c \right] = e^{-t/RC} \left[\frac{E_0}{R} \frac{e^{t/RC} \left(\frac{1}{RC} \sin \omega t - \omega \cos \omega t \right)}{\frac{1}{R^2 C^2} + \omega^2} + c \right] \\ &= E_0 C \left[\frac{(\sin \omega t - RC \omega \cos \omega t)}{1 + R^2 C^2 \omega^2} \right] + c e^{-t/RC} \end{aligned}$$

Using the initial condition $Q(0) = 0$, this equation gives: $c = \frac{E_0 C^2 R \omega}{1 + R^2 C^2 \omega^2}$. Thus we get the solution as,

$$Q(t) = \frac{E_0 C}{1 + R^2 C^2 \omega^2} [\sin \omega t - RC \omega (\cos \omega t - e^{-t/RC})]. \quad (4.9.15)$$

Example 4.9.5 (LR Circuit)

A 12-volt battery is connected to a series circuit in which the inductance is $\frac{1}{4}$ henry and the resistance is 20 ohms. Determine the current I if the initial current is zero.

Solution

We are given: $E(t) = 12$ volts, $L = \frac{1}{4}$ henry, $R = 20$ ohms, and $I(0) = 0$. Thus using Eqn.(4.8.1) we get,

$$\frac{1}{4} \frac{dI}{dt} + 20 I = 12, \text{ or } dI/dt + 80 I = 48.$$

It is a linear differential equation of first order with integrating factor, e^{80t} . Thus we get the solution of the above equation as,

$$I = e^{-80t} [\int e^{80t} 48 dt + c] = e^{-80t} \left[\frac{3}{5} e^{80t} + c \right] = \frac{6}{5} + c e^{-80t}.$$

Since $I(0) = 0$, so the above equation gives: $c = -3/5$. Therefore, we get,

$$I(t) = \frac{3}{5} [1 - e^{-80t}], \quad (4.9.16)$$

which gives the required current at any time t .

In the solution (4.9.16), the first term on the right hand side gives the **steady state solution**, while the second term gives the **transient solution**. We see that as $t \rightarrow \infty$, the transient solution term approaches zero. Therefore, for large values of time t , the flow of current in the circuit is governed by the Ohm's law, $E = IR$, that gives $I = E/R = 3/5$.

Note.

LRC-circuits containing three kinds of components (resistor, inductor and capacitor) lead to second order differential equations which will be dealt with later.

EXERCISE – 4.6

[LR & RC Circuits]

Note: Answers are given against the respective questions within square brackets.

LR-Circuits

- 1 Solve the differential equation for the LR-series circuit (Eqn. (4.9.1)), taking $E = E_0$, satisfying the initial conditions (i) $I(0) = 0.5 E_0/R$, (ii) $I(0) = E_0/R$, (iii) $I(0) = 2E_0/R$. Also sketch the corresponding curves.

$$[(i) (E_0/R)(1 - 0.5 e^{-tR/L}), (ii) E_0/R, (iii) (E_0/R)(1 + e^{-tR/L})]$$

- 2 In Example 4.9.1, taking $E(t) = E_0$ (constant), $R=20$ Ohms, $L = 0.03$ millihenry ($= 3 \times 10^{-5}$ henry) and $I(0) = 0$, find the time when the current reaches 99.9% of its final value.

$$[t = (L/R) \ln 1000 = (3 \times 10^{-5}/20) \ln 1000 = 1.036 \times 10^{-5} \text{ sec}]$$

3. In an LR-circuit, $R = 12$ ohms, $L = 0.02$ millihenry, $E = E_0 = \text{constant}$, and $I(0) = 0$. How long will it take the current $I(t)$ to reach practically (theoretically 99.9% of) its final value?

$$[1.15 \times 10^{-5} \text{ sec}]$$

4. In Problem 2 above, will the time increase or decrease if we change $I(0) = 0$ to some positive value less than E_0/R ? **[Time will decrease]**

5. What L should be chosen in an LR-circuit with $R = 100$ ohms for Eqn. (4.9.4) to reach 99.9% of its final value at $t = 0.01$ sec? **[$L = 1/(\ln 1000) = 0.145$ henry]**
6. What L should be chosen in Eqn. (4.9.1) with $E = 100$ volts and $R = 1000$ ohms if we want the current to grow from 0 to 25% of its final value within 10^{-4} sec?
[$L = 0.348$ henry]
7. In Example 4.9.1 let $R = 100$ ohms, $L = 2.5$ henrys, $E = 110$ volts and $I(0) = 0$. Find the time necessary for the current to rise from 0 to 0.6 ampere. **[$t = 0.02$ sec]**
8. Derive the steady state solution for Eqn.(4.9.8) by substituting $I_p = A \cos \omega t + B \sin \omega t$ into Eqn.(4.9.1) with $E(t) = E_0 \sin \omega t$ and determining A and B by equating the cosine and sine terms in the resulting equation. (This avoids integration by parts)
[$A = \frac{-\omega L E_0}{R^2 + \omega^2 L^2}$, $B = \frac{E_0 R}{R^2 + \omega^2 L^2}$, $I_p = \frac{E_0}{R^2 + \omega^2 L^2} \{R \sin \omega t - \omega L \cos \omega t\}$]
9. How does the phase angle α in Eqn.(4.9.9) depend on L ? Is this physically understandable?
[α increases with L , indicating that L has a retarding effect (an inertia effect)]
10. Solve Eqn. (4.9.1) with $E(t) = e^{-t}$ when (i) $R \neq L$, (ii) $R = L$.
[(i) $I(t) = c e^{-tR/L} + e^{-t}/(R-L)$, (ii) $I(t) = (c + t/L) e^{-t}$]
11. Solve Eqn. (4.9.1) with $E(t) = t$. **[$I(t) = c e^{-tR/L} + t/R - L/R^2$]**
12. Find the current in LR-circuit with $R=1$ ohm, $L = 1$ henry, and $E(t) = 1$ if $0 \leq t \leq 3$ sec, $E(t) = 0$ if $t > 3$ sec, $I(0) = \frac{1}{2}$ ampere. [Hint: Use the concept that $I(t)$ must be continuous at $t = 3$].
[$I = 1 - e^{-t}/2$ if $0 \leq t \leq 3$, $I = I(3) e^{-(t-3)}$, where $I(3) = 1 - e^{-3}/2 = 0.975$]
13. If $L = 10$ in Problem 3, what R should we choose for $I(t)$ to reach practically (theoretically 99% of) its final value after 1 sec? **[46 ohms]**

- 14 A 30 volt electromotive force is applied to an LR-series circuit in which the inductance is 0.1 henry and the resistance is 50 ohms. Find the current $I(t)$ if $I(0) = 0$. Determine the current as $t \rightarrow \infty$. **[$I(t) = \frac{3}{5} (1 - e^{-500t})$, $I \rightarrow \frac{3}{5}$ as $t \rightarrow \infty$]**
- 15 An electromotive force: $E(t) = 120$ ($0 \leq t \leq 20$), $E(t) = 0$ ($t > 20$), is applied to an LR series circuit in which the inductance is 20 henrys and the resistance is 2 ohms. Find the current $I(t)$ if $I(0) = 0$.
[$I(t) = 60 (1 - e^{-t/10})$ for $0 \leq t \leq 20$, and $I(t) = 60 (e^2 - 1) e^{-t/10}$ for $t > 20$]

RC-Circuits

- 16 Suppose that an RC series circuit has a variable resistor. If the resistance at any time t is given by $R = k_1 + k_2 t$, where k_1 and k_2 , both greater than zero, are known constants, then Eqn.(4.9.3) becomes,

$$(k_1 + k_2 t) \frac{dQ}{dt} + \frac{1}{C} Q = E(t).$$

Also show that if $E(t) = E_0$ and $Q(0) = Q_0$, then,

$$Q(t) = E_0 C + (Q_0 - E_0 C) \left(\frac{k_1}{k_1 + k_2 t} \right)^{1/Ck_2}.$$

- 17 Solve Eqn. (4.9.3), i.e., $R \frac{dQ}{dt} + \frac{1}{C} Q = E(t)$, with $E(t) = 0$, assuming $Q(0) = Q_0$. Find the time when the capacitor has lost 99.9% of its initial charge.

$$[Q = Q_0 e^{-t/RC} \text{ coulomb; } t = RC \ln 1000 = 6.908 RC \text{ sec}]$$

- 18 In Eqn. (4.9.3), let $R = 20$ ohms, $C = 0.01$ farad, and $E(t)$ be exponentially decaying, say, $E(t) = 60 e^{-2t}$ volts. Assuming $Q(0) = 0$, find and sketch $Q(t)$. Also determine the time when $Q(t)$ reaches a maximum and that maximum charge.

$$[Q(t) = e^{-2t} - e^{-5t}, t = 0.3054 \text{ sec, } Q_{\max} = 0.3257 \text{ coulomb}]$$

- 19 A capacitor ($C = 0.1$ farad) in series with a resistor ($R = 200$ ohms) is charged from a source ($E = E_0 = 12$ volts). Find the voltage $V(t)$ on the capacitor, assuming that at $t = 0$ the capacitor is completely uncharged.

$$[RQ' + Q/C = E_0, Q(t) = E_0 C (1 - e^{-t/RC}), V(t) = Q(t)/C = 12 (1 - e^{-0.05t})]$$

- 20 Find the current $I(t)$ in an RC-circuit assuming that $E = 100$ volts, $C = 0.25$ farad, R is variable according to $R = (100-t)$ ohms when $0 \leq t \leq 100$ sec, $R = 0$ when $t > 100$ sec, and $I(0) = 1$ ampere.

$$I(t) = 10^{-6} (100-t)^3 \text{ if } 0 \leq t \leq 100, \quad I(t) = 0 \text{ if } t > 100]$$

- 21 Solve Eqn. (4.9.3) when $R = 500$ ohms, $C = 10^{-3}$ farad, $E(t) = 1 - e^{-t}$ volt.

$$[Q(t) = c e^{-2t} + 0.001 (1 - 2 e^{-t})]$$

- 22 Find the steady-state solution of Eqn. (4.9.3) when $R = 50$ ohms, $C = 0.04$ farad, and $E(t)$ equals: (i) $110 \cos 314t$, (ii) $50 e^{-t} + 1012 \sin \pi t$.

$$[(i) \quad Q = 0.000011 \cos 314t + 0.007006 \sin 314t, \quad (ii) \quad Q = \sin \pi t - 2\pi \cos \pi t]$$

- 23 A 100 volts electromotive force is applied to an RC series circuit in which the resistance is 200 ohms and the capacitance is 10^{-4} farad. Find the charge $Q(t)$ on the capacitor if $Q(0) = 0$. Find the current $I(t)$.

$$[Q(t) = 0.01 (1 - e^{-50t}); \quad I(t) = 0.5 e^{-50t}]$$

- 24 A 200 volts electromotive force is applied to an RC series circuit in which the resistance is 1000 ohms and the capacitance is 5×10^{-6} farad. Find the charge $Q(t)$ on the capacitor if $I(0) = 0.4$. Determine the charge and current at $t = 0.005$ sec. Also determine the charge as $t \rightarrow \infty$.

$$[Q(t) = 0.001 - 0.002 e^{-200t}; \quad Q(0.005) = 0.003 \text{ coulomb},$$

$$I(0.005) = 0.1472 \text{ amperes; as } t \rightarrow \infty, Q(t) \rightarrow 1/1000]$$

- 25 Show that if the initial charge in a capacitor in an RC-circuit is $Q(0)$, the initial current in this circuit is $I(0) = E(0)/R - Q(0)/RC$.

- 26 Show that if $E(t)$ in an RC-circuit has a jump of magnitude J at $t = a$, then the current $I(t)$ in this circuit has a jump of magnitude J/R at $t = a$.

[Since the charge on the capacitor cannot change abruptly we must have $J =$ jump of RI , hence jump of $I = J/R$]

- 27 Find the current $I(t)$ in an RC-circuit, assuming $R = 1$ ohm, $C = 1$ farad, zero initial charge on the capacitor, i.e., $I(0) = 0$, $Q(0) = 0$,

(i) $E = t$ when $0 \leq t \leq 1$, and $E = 1$ when $t > 1$.

$$[I(t) = 1 - e^{-t} \text{ when } 0 \leq t \leq 1, \text{ and } I(t) = (e - 1) e^{-t} \text{ when } t > 1]$$

(ii) $E = t + 1$ when $0 \leq t \leq a$ and $E = 0$ when $t > a$.

$$[I(t) = 1 \text{ when } 0 \leq t \leq a, \text{ and } I(t) = -a e^{a-t} \text{ when } t > a]$$

(iii) $E = t + 1$ when $0 \leq t \leq a$, and $E = a + 1$ when $t > a$.

$$[I(t) = 1 \text{ if } 0 \leq t \leq a, \text{ and } I(t) = e^{-(t-a)} \text{ if } t > a]$$

- 28 **(Heart Pacemaker)** A heart pacemaker as shown in Fig. 4.9.6 consists of a battery of voltage E_0 , a capacitor of capacitance C , the heart as a resistor of resistance R , and a switch that is periodically moved from P (charging period $t_1 < t < t_2$ of the capacitor) to Q (discharging period $t_2 < t < t_3$ during which the capacitor sends an electric stimulus to the heart). Find the current during the discharging period, assuming that the charging period is such that the charge on the capacitor is 99% of its maximum possible value.

$$[I = I_0 e^{-(t-t_2)/RC}, \text{ with } I_0 = -0.99 E_0/R]$$

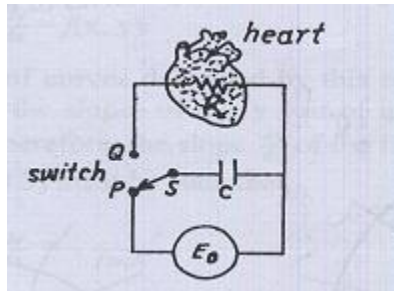


Fig. 4.9.6. Heart Pacemaker

- 29 A heart pacemaker, as shown in Fig. 4.9.6 consists of a battery, a capacitor, and the heart as a resistor. When the switch S is at P , the capacitor charges; when S is at Q , the capacitor discharges, sending an electrical stimulus to the heart. During this time the voltage E , applied to the heart is given by,

$$\frac{dE}{dt} = -\frac{1}{RC} E, \quad (t_1 < t < t_2)$$

where R and C are constants. Determine $E(t)$ if $E(t_1) = E_0$. (Of course the opening and closing of switch are periodic in time, to simulate the natural heartbeat).

$$[E(t) = E_0 e^{-(t-t_1)/RC}]$$

4.10 Orthogonal Trajectories

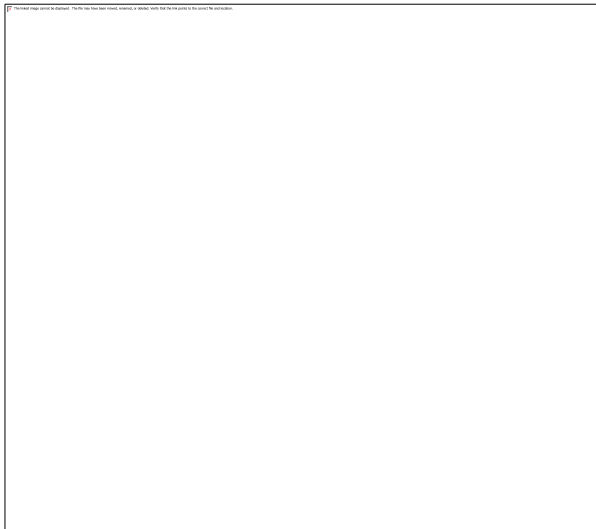
An interesting problem that often occurs in mathematics, physics and engineering is to find a family curves, say,

$$f_1(x, y, c_1) = 0, \quad (4.10.1)$$

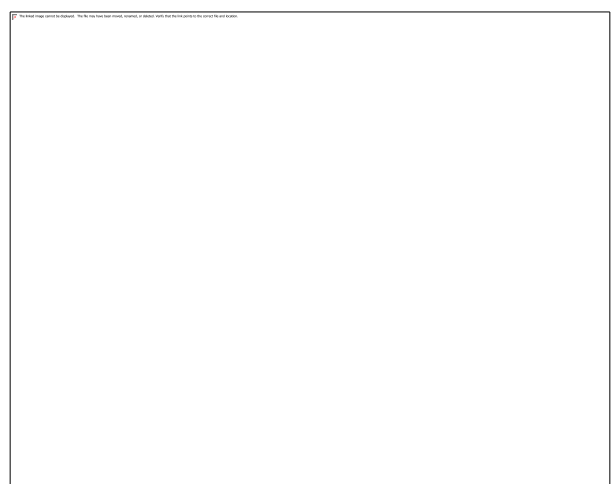
(c_1 being a parameter) that intersects another family of curves, say,

$$f_2(x, y, c_2) = 0, \quad (4.10.2)$$

(c_2 being a parameter) orthogonally, i.e., at right angles, at each point. Then both these families of curves are called ***orthogonal trajectories*** of each other (see Fig 4.10.1).



(i)



(ii)

Fig 4.10.1: Orthogonal Trajectories of the family of (i) Circles & St. Lines, (ii) Ellipses & Hyperbolas

Some example of **orthogonal trajectories** are given as under,

- a. Lines of force in an electric field are orthogonal to equi-potential curves (i.e., curves of constant voltage) and conversely.
- b. The meridians on the globe are orthogonal to the parallels.
- c. The curves of steepest descent on a map are orthogonal to the contour lines.
- d. In heat conduction a family of curves represents lines of constant temperature (called isotherms), and their orthogonal trajectories represent lines of heat flow. (Many other examples arise in the fields of physics and engineering including heat conduction, fluid flow, etc)
- e. In two-dimensional fluid dynamics the curves followed by fluid particles (called streamlines) are orthogonal to the associated equi-potential lines, (along which a function called the “fluid potential” is constant).
- f. In two dimensional electromagnetic theory, family of curves describing lines of constant electric potential (equi-potential lines) are orthogonal to flux lines.

In order to find orthogonal trajectories of the family,

$$F(x, y, c) = 0 \quad (4.10.3)$$

we differentiate it w.r.t. x and then eliminate c between this equation and the equation obtained by differentiating it. Accordingly, we obtain a differential equation of the form,

$$\frac{dy}{dx} = f(x, y). \quad (4.10.4)$$

For the family of curves described by this equation to be orthogonal to another family, the products of the slopes of every pair of intersecting curves must be equal to -1 (i.e., $m_1 m_2 = -1$). Therefore, the slope $\frac{dy}{dx}$ of the family of curves that are mutually orthogonal to those of (4.10.3) must be such that,

$$\frac{dy}{dx} = -\frac{1}{f(x,y)} \quad (4.10.5)$$

This is the differential equation of the required family of orthogonal trajectories. Therefore, to find the orthogonal trajectories we follow the steps as given below:

Step-1. Given a family of curves $F(x,y,c) = 0$. We differentiate it and eliminate c between the given equation and the one obtained by differentiating it. Thus we obtain an equation of the form, $\frac{dy}{dx} = f(x,y)$.

Step-2. We find the orthogonal trajectories by solving the differential equation,

$$\frac{dy}{dx} = -\frac{1}{f(x,y)}$$

This equation can usually be solved by the method of separation of variables (see Sec 3.2).

Example 4.10.1

Find the orthogonal trajectories of all parabolas with vertices at the origin and foci on the x -axis.

Solution

Step-1.

The parabolas with vertices at the origin and focus on the x -axis are given by the equation,

$$y^2 = 4ax, \quad (4.10.6)$$

where 'a' is a parameter. Differentiating it we obtain, $\frac{dy}{dx} = \frac{2a}{y}$. Eliminating the parameter 'a' between this equation and the given equation (4.10.6) we get,

$$\frac{dy}{dx} = f(x, y) = \frac{y}{2x} \quad (4.10.7)$$

Step-2.

Therefore, the orthogonal trajectories of the family (4.10.6) must satisfy the equation,

$$\frac{dy}{dx} = -\frac{1}{f(x,y)} = -\frac{2x}{y}. \quad (4.10.8)$$

Solving it we obtain,

$$2x^2 + y^2 = b^2, \quad (4.10.9)$$

where b is an arbitrary constant. Therefore, the orthogonal trajectories of the given family of parabolas (4.10.6) are the ellipses (4.10.9).

Example 4.10.2

Find the orthogonal trajectories of the family of circles,

$$x^2 + y^2 = r^2, \quad (4.10.10)$$

where r (radius) is a parameter.

Solution**Step-1**

Differentiating Eqn.(4.10.10) w.r.t. x we get,

$$\frac{dy}{dx} = -\frac{x}{y} \quad (4.10.11)$$

Step-2

The orthogonal trajectories of the given family of curves must satisfy the differential equation,

$$\frac{dy}{dx} = \frac{y}{x}. \quad (4.10.12)$$

Solving this equation we, get,

$$y = kx, \quad (4.10.13)$$

where k is an arbitrary constant. Therefore, the orthogonal trajectories of the given family of circles (4.10.10) are the family of hyperboles (4.10.13).

Example 4.10.3

Find the orthogonal trajectories of the hyperboles,

$$x^2 - y^2 = c^2 \quad (4.10.14)$$

Solution**Step-1**

On differentiating (4.10.14) w.r.t. x we obtain,

$$\frac{dy}{dx} = \frac{x}{y} \quad (4.10.15)$$

Step –2

The orthogonal trajectories of the given family of curves must satisfy the differential equation,

$$\frac{dy}{dx} = -\frac{y}{x} \quad (4.10.16)$$

On solving it we get,

$$xy = k, \quad (4.10.17)$$

where k is an arbitrary constant. Thus, the orthogonal trajectories of the given family of hyperboles (4.10.14) are the family of hyperboles (4.10.17).

EXERCISE 4.7
(Orthogonal Trajectories)

Note: Answers are given against the respective questions within square brackets.

Find the orthogonal trajectories of the following families of curves. Also draw the figures:

1. $x - 4y = c$ [$4x + y = k$]
2. Circles through the origin with centres on the x-axis
[**Circles through the origin with centres on the y – axis**]
3. Straight lines with slope and y-intercept equal. [$(x + 1)^2 + y^2 = r^2$]
4. $x^2 + y^2 = r^2$ [$y = kx$]
5. $x^2 - y^2 = c$ [$xy = k$]
6. $y^2 = cx^3$ [$2x^2 + 3y^2 = k^2$]
7. $e^x + e^{-y} = c$ [$e^y - e^{-x} = k$]
8. $y = c (\sec x + \tan x)$ [$y^2 = 2(k - \sin x)$]
9. $x^3 = 3(y - c)$ [$x(y - k) = 1$]
10. $x = c e^{y^2}$ [$y = c e^{-x^2}$]
11. Ellipses with centres at (0, 0) and two vertices at (1, 0) and (-1, 0).
[$x^2 + y^2 = 2 \ln |kx|$]
12. $x^2 - y^2 = cx$ [$y(y^2 + 3x^2) = k$]